

## ESSENTIAL MATHEMATICS GRADE 10: VOLUME AND CAPACITY

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.7 (8 Lessons)*

| SUB-STRAND OVERVIEW |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| Grade Level         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Subject             | Essential Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Strand              | Strand 2.0: Measurements and Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Sub-Strand          | Sub-Strand 2.7: Volume and Capacity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Total Duration      | 8 lessons × 40 minutes = 320 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Sub-Strand Content  | <ul style="list-style-type: none"> <li>– Volume and capacity of cones: deriving and applying the formula <math>V = \frac{1}{3}\pi r^2 h</math>; relating to the volume of a cylinder of equal base and height</li> <li>– Volume and capacity of pyramids (square and rectangular base): deriving and applying <math>V = \frac{1}{3} \times \text{base area} \times \text{height}</math></li> <li>– Volume and capacity of frustums of cones: applying <math>V = \frac{1}{3}\pi h(R^2 + Rr + r^2)</math> using dimensions of the original cone and the removed top cone</li> <li>– Volume and capacity of frustums of pyramids: applying <math>V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})</math> for square and rectangular bases</li> <li>– Relationship between volume units and capacity units (<math>\text{cm}^3 \leftrightarrow \text{mL}</math>, <math>\text{m}^3 \leftrightarrow \text{L}</math>; <math>1 \text{ L} = 1\,000 \text{ cm}^3</math>)</li> <li>– Reading and interpreting measurements of real containers (buckets, grain stores, water tanks, cooking pots) shaped as cones, pyramids and frustums</li> <li>– Selecting and applying appropriate formulae to solve real-life volume and capacity problems</li> <li>– Project: making and measuring model containers shaped as frustums, pyramids and cones from locally available materials</li> </ul> |
| Learning Outcomes   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>determine the volume and capacity of cones and pyramids,</li> <li>compute volume and capacity of a frustum of a cone,</li> <li>calculate the volume and capacity of a frustum of a pyramid,</li> <li>apply volume and capacity in real-life situations,</li> <li>appreciate volumes and capacities in real life situations.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Core Competencies   | <ul style="list-style-type: none"> <li>– Critical Thinking and Problem Solving: the learner acquires open-mindedness and creativity skills when discussing with peers and working out the volume and capacity of containers shaped as cones, pyramids and frustums; learners must choose and justify the correct formula for each shape encountered.</li> <li>– Creativity and Imagination: the learner develops connection and networking skills when deriving the volume of frustums by subtracting a smaller cone or pyramid from a larger one, and when constructing physical models of these solids from locally available materials such as cardboard, old packaging and clay.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Core Values         | <ul style="list-style-type: none"> <li>– Unity: the learner displays tolerance of different opinions while working with peers when constructing models of frustums as learning resources, sharing materials equitably and dividing tasks fairly within the group.</li> <li>– Respect: the learner listens to others' opinions while discussing appropriate formulae for working out volume and capacity of containers, and acknowledges contributions of group members during presentations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| <b>Science &amp; Engineering Practices</b>        | <ul style="list-style-type: none"> <li>– Asking Questions and Defining Problems: learners pose questions about why different containers (buckets, grain stores, water tanks) hold different amounts even when they look similar in height, initiating the mathematical investigation of volume and capacity.</li> <li>– Developing and Using Models: learners build physical models of cones, pyramids and frustums to visualise the relationship between a frustum and its parent solid, then use mathematical formulae as abstract models to predict capacity.</li> <li>– Using Mathematics and Computational Thinking: learners select and apply the appropriate volume formula, substitute measured dimensions, and convert between volume units (<math>\text{cm}^3</math>, <math>\text{m}^3</math>) and capacity units (mL, L) to solve real-life container problems.</li> <li>– Constructing Explanations: learners explain why <math>V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})</math> by referencing their physical models and measurements, justifying each step of their working.</li> </ul>                                                                       |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Environment and Technology: the learner uses appropriate containers from the local environment — such as plastic buckets used for carrying water from the borehole, metal grain-storage bins common in Rift Valley farms, and cooking pots used for ugali — to work out the capacity of cones, pyramids and their frustums, promoting environmental awareness and resourcefulness.</li> <li>– Education for Sustainable Development: by calculating the capacity of water-harvesting tanks and grain-storage structures shaped as frustums, learners connect mathematics to sustainable food and water management practices relevant to Kenyan farming communities.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Civil and Structural Engineering: calculating volumes of conical and pyramidal roof structures, water tanks and grain silos used in construction projects across Kenya.</li> <li>– Architecture and Interior Design: determining capacity of decorative containers, lamp shades and furniture components with frustum or pyramidal shapes.</li> <li>– Agriculture and Food Science: estimating grain storage capacity of pyramidal or frustum-shaped stores; calculating volumes of conical heaps of maize or beans at market stalls in Wakulima Market, Nairobi.</li> <li>– Manufacturing and Packaging: designing and sizing packaging (e.g., paper funnels, ice-cream cones, juice cartons) for the food and beverage industry in Kenyan factories.</li> <li>– Hospitality and Catering: working out capacities of cooking vessels and serving containers in hotels and restaurants along the Kenyan coast and urban centres such as Kisumu and Mombasa.</li> <li>– Water Resources Management: computing the volume of water stored in conical or frustum-shaped tanks for community water projects in arid and semi-arid areas of Kenya.</li> </ul> |
| <b>Focus for Lessons</b>                          | This unit develops learners' ability to calculate the volume and capacity of cones, pyramids and their frustums, rooting every concept in Kenyan everyday objects such as water-carrying buckets, grain-storage bins and cooking pots. Learners move from establishing the cone–cylinder relationship through hands-on measurement of real containers to applying and justifying formulae for frustums of both cones and pyramids. The unit culminates in a project in which learners design, build and document model containers, connecting geometry to sustainable resource management in the Kenyan context.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Driving Question / Key Inquiry Questions</b>   | <p>DRIVING QUESTION: Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?</p> <p>KICD KEY INQUIRY QUESTION:</p> <p>1. Why is the knowledge of volume and capacity useful?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Anchoring Phenomenon</b>                       | A group of farmers in Kericho have three different containers they use to store harvested tea leaves before collection: a tall cylindrical drum, a conical heap-cover made from woven sisal, and a metal bucket (frustum of a cone) used for carrying leaves to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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|                         | <p>the weighing station. All three containers have roughly the same height and the same opening diameter, yet when filled with water the cylindrical drum holds exactly three times as much water as the conical cover, and the metal bucket holds noticeably more water than the cone but less than the cylinder. Learners observe these three containers side by side, filled with water, and are puzzled: why do shapes that look so similar in size hold such dramatically different amounts? This anchoring phenomenon drives the entire unit — learners must unpack the mathematical relationship between a cone and a cylinder, then extend that reasoning to pyramids and frustums, to explain and predict the capacities of the containers used every day by Kenyan tea farmers.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Storyline Thread</b> | <p>Lesson 1 — Predict &amp; Anchor Phenomenon: Learners examine the three Kericho tea-farmer containers (or physical models/images thereof). They predict which holds the most water, record estimates, and articulate what they wonder. The driving question is introduced and posted on the Driving Question Board. Learners review the relationship between volume units (<math>\text{cm}^3</math>, <math>\text{m}^3</math>) and capacity units (mL, L) as prerequisite knowledge.</p> <p>Lesson 2 — Cone vs. Cylinder (Observe &amp; Explain): Using a cone and cylinder of identical base radius and height (models or demonstration containers), learners fill the cone with water and pour it into the cylinder three times to observe that <math>V(\text{cylinder}) = 3 \times V(\text{cone})</math>. They derive the cone volume formula <math>V = \frac{1}{3}\pi r^2 h</math>, apply it to the Kericho conical cover, and calculate its capacity in litres. Learners update the DQB with new understanding.</p> <p>Lesson 3 — Volume of Pyramids (Observe &amp; Explain): Learners use models of a square-based pyramid and a cuboid of equal base and height to establish <math>V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}</math>. They apply the formula to pyramidal grain-store models, computing volumes and converting to capacity. They compare results with their Lesson 1 predictions and refine the DQB.</p> <p>Lesson 4 — Introducing Frustums of Cones (Explain &amp; Model Building): Learners measure a real frustum-shaped bucket (top radius <math>R</math>, bottom radius <math>r</math>, height <math>h</math>) and the dimensions of the full cone from which it could be cut. They derive <math>V(\text{frustum of cone}) = V(\text{large cone}) - V(\text{small cone}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)</math> step by step. They calculate the bucket's capacity and compare it to the cylindrical drum, resolving the anchoring phenomenon.</p> <p>Lesson 5 — Volume of Frustums of Pyramids (Explain &amp; Model Building): Learners extend the frustum concept to square and rectangular pyramidal frustums using <math>V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})</math>. They apply the formula to a pyramidal grain-storage bin model, compute capacity in litres and kilograms of maize (using density), and update their cumulative models on the DQB.</p> <p>Lesson 6 — Problem Solving with Real Containers (Apply &amp; DQB Update): Learners work in groups on a set of Kenyan-context word problems: sizing a conical water tank for a homestead in Turkana, calculating the capacity of a frustum-shaped cooking pot used in a githeri recipe for 100 people, and determining how many frustum-shaped cups fill a cylindrical storage drum. Groups present solutions and justify formula choices; the DQB is updated with consolidated understanding.</p> <p>Lesson 7 — Project: Model Making &amp; Measurement: Learners design and construct model containers shaped as a frustum of a cone, a frustum of a pyramid and a cone using locally available materials (cardboard from cereal boxes, old packaging). They measure dimensions, calculate theoretical volumes and capacities, then fill models with water or sand to verify experimentally. Discrepancies between theoretical and measured values are discussed. Photographs and video clips are taken for documentation.</p> <p>Lesson 8 — Model Building Culmination &amp; Assessment: Groups present their project models, share calculated and measured capacities, and explain sources of error. The class revisits the anchoring phenomenon — the Kericho farmers' containers — and constructs a final annotated diagram showing the volume relationships between cone, cylinder, pyramid and frustum. A written class activity assesses all five specific learning outcomes. The DQB is completed and learners articulate what they now know</p> |

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|                             | versus what they predicted in Lesson 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– A mandazi vendor in Mombasa uses a conical paper bag and a frustum-shaped plastic cup to pack mandazi for customers; the vendor claims both hold the same number of mandazi, but customers dispute this — learners investigate whether the volumes are truly equal given the same slant height.</li> <li>– A water-harvesting project near Lake Turkana installs frustum-shaped underground cisterns; community members want to know how many 20-litre jerricans can be filled from a cistern of given top diameter, bottom diameter and depth — learners calculate the capacity and convert to jerricans.</li> <li>– A Nairobi school's mathematics club builds a square pyramidal display stand for their science fair; the teacher asks them to fill it with sand to find its volume experimentally, then compare the result to the formula — learners reconcile measured and calculated values.</li> </ul> |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (40 minutes): The Kericho Container Puzzle: Anchoring the Phenomenon & Launching the Driving Question

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | To anchor learners in the phenomenon of three Kericho tea-farmer containers (cylinder, cone, frustum) that look the same size yet hold dramatically different amounts of water, and to activate prerequisite knowledge of volume and capacity units.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Knowledge</b>            | Learners recall and articulate the relationship between volume units ( $\text{cm}^3$ , $\text{m}^3$ ) and capacity units (mL, L); learners identify the names and key features (height, base radius, slant) of a cylinder, cone, and frustum (bucket) as seen in the Kericho tea-farming context.                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Skills</b>               | Learners make and record quantitative predictions; learners estimate and compare capacities of the three containers; learners formulate questions about the phenomenon and post them on the Driving Question Board.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Attitudes</b>            | Learners appreciate that mathematics embedded in everyday Kenyan farm tools can reveal surprising and useful truths; learners demonstrate curiosity, honesty in recording predictions, and respect for peers' differing estimates.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Purpose in Storyline</b> | This is the opening lesson of the unit. Learners encounter the anchoring phenomenon — three containers used by Kericho tea farmers that appear similar in size but hold strikingly different volumes of water. By predicting, observing side-by-side images or physical models, and articulating their wonderings, learners generate the intellectual need that will drive the entire unit. The Driving Question Board is opened and the unit's driving question is posted. Prerequisite knowledge of $\text{cm}^3/\text{m}^3 \leftrightarrow \text{mL}/\text{L}$ conversions is reviewed so learners have the language to quantify their predictions in subsequent lessons. |
| <b>Safety Notes</b>         | If physical containers with water are used, place a plastic sheet or basin under the demonstration area to prevent slipping on wet floors. Learners handle models carefully; no sharp cutting tools are used in this lesson.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### B. LESSON OVERVIEW

Learners are introduced to the unit's anchoring phenomenon: three containers used by Kericho tea farmers — a tall cylindrical drum, a conical sisal heap-cover, and a metal frustum-shaped bucket — that have nearly the same height and opening diameter yet hold dramatically different amounts of water. Working individually first, then in pairs, learners study images or physical models, predict which container holds the most and least water, record numerical estimates in litres, and share their reasoning. The class articulates what surprises them and what they wonder, generating questions that are posted on the newly opened Driving Question Board. The teacher then facilitates a brief prerequisite review connecting  $\text{cm}^3$  to mL and  $\text{m}^3$  to L, equipping learners with the unit language they will need throughout. The lesson closes with the formal posting of the unit Driving Question.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                               | Learner Experience                                                                                                                      | Teacher Moves                                                                                                                       | Sensemaking Strategy                                                                                                                 | Formative Assessment Strategy                                                                                                     |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Visually Comparing the</a> | Each learner receives a printed A5 observation sheet showing three side-by-side labelled diagrams of the Kericho tea-farmer containers: | Distribute observation sheets face-down. Say: 'Before we do any mathematics today, I want you to trust your eyes and your instincts | Think-Pair-Share with ranked prediction: Learners first commit to a personal prediction in writing (individual accountability), then | Teacher scans observation sheets during the silent phase to check: (a) all learners have attempted a ranking — redirect any blank |

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| <p><a href="#">Capacity and Volume of Three Containers (Metric Units)</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/><a href="#">Volume of the Cone (PLIX)</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p>                                                                      | <p>Container A (cylinder), Container B (cone, apex down), and Container C (frustum/bucket). Dimensions shown: all three have a height of approximately 30 cm and a top opening diameter of approximately 20 cm. Learners study the diagrams silently for 2 minutes, then individually write responses to three prompts on their sheet: (1) Rank the containers from most to least water held — write your ranking and explain your reasoning in one sentence. (2) Estimate in litres how much water each container holds when filled to the brim. (3) Write one thing you notice and one thing you wonder about these containers. Learners then share their ranking and one estimate with a partner using a Think-Pair-Share, with each partner taking 30 seconds to explain their reasoning before hearing the other's view.</p> | <p>— just like a tea farmer in Kericho would when choosing the right container for the job.' Flip sheets over on count of three. Allow 2 minutes of silent individual thinking — circulate and observe without commenting. After 2 minutes say: 'Now write your ranking — most to least — and your estimates in litres. You have 90 seconds.' After writing, say: 'Turn to your partner. Partner A, share your ranking and your reason — you have 30 seconds. Then Partner B responds.' After both share, cold-call 4 pairs (choose 1 pair who ranked <math>C &gt; A &gt; B</math>, 1 pair who ranked <math>A &gt; C &gt; B</math>, and 2 pairs with other rankings) using: 'Tell the class your ranking and the single most important reason behind it.' After each response, ask the class: 'Does anyone have a different reason for the same ranking?' Wait 10 seconds before calling on a volunteer. Record all three distinct rankings on the board in a simple tally. Do NOT confirm or deny any prediction — say: 'All of these rankings are on the board. By the end of this unit, every one of you will be able to prove which ranking is correct using mathematics.'</p> | <p>encounter a peer's potentially different view (cognitive conflict), which surfaces the phenomenon's puzzle before any instruction occurs. Recording all rankings publicly normalises uncertainty and establishes that the phenomenon is genuinely surprising.</p>                                                                                                                                                      | <p>sheets with 'Just write what your eyes tell you, there is no wrong answer yet'; (b) estimates are expressed as numbers with a unit — if a learner writes only 'big' or 'medium', prompt: 'Can you give me a number in litres or millilitres?' Cold-call responses reveal whether learners are reasoning from height alone, from the visual width, or from a vague sense of shape — this diagnostic informs which misconceptions to address in later lessons.</p> |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Metric system: units of volume</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/><a href="#">Volume of Cylinder: Big Can</a></p> | <p>The teacher projects (or displays on printed A3 posters) a sequence of three photographs or diagrams taken from a Kericho tea farm: (i) the cylindrical drum alone, filled to the brim with water coloured with blue food dye; (ii) the conical sisal cover, filled to the brim — the same blue water is poured in and the level is visibly lower than expected; (iii) the frustum metal bucket, filled to the brim — water level is between the cone and the</p>                                                                                                                                                                                                                                                                                                                                                              | <p>Display images one at a time. Before revealing the fourth image say: 'I am about to show you the actual measured capacities. Before I do — look at your estimate. On your sheet, circle it.' Reveal the fourth image. Allow 30 seconds of silent reaction. Then say: 'Update your observation sheet — write the actual values and put a tick or cross next to your ranking.' Give 1 minute. Then say: 'In your groups of three, discuss</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Discrepant-event observation: The gap between learners' predictions and the actual capacities (especially the striking 3:1 cylinder-to-cone ratio) creates productive cognitive dissonance. This is the engine of inquiry — learners now have a genuine need to know why. The group hypothesis step moves learners from passive surprise to active preliminary explanation, practising the science and engineering</p> | <p>Listen to group hypothesis sentences. Observable indicators of understanding: (a) groups that connect the cone's narrowing shape to less volume are ready for formal derivation (Lesson 2); (b) groups that only say 'the cone is smaller' without referencing shape change need the cylinder-cone comparison demonstration in Lesson 2 to be more hands-on; (c) groups that are fixated solely on the opening diameter rather than</p>                          |



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| <p><a href="#">of Soup (PLIX)</a><br/>Source: CK-12<br/><a href="#">Search ARES</a><br/>for similar readings</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>cylinder. A fourth image shows all three containers side by side, each filled to the brim, with measuring jugs placed next to each showing the actual capacities: A (cylinder) <math>\approx</math> 9.4 L, B (cone) <math>\approx</math> 3.1 L, C (frustum/bucket) <math>\approx</math> 6.5 L. Learners record the actual capacities on their observation sheet, note whether their ranking was correct, and update their 'I wonder' statement based on what they now see. In groups of three, learners discuss: 'The cylinder holds almost exactly three times as much as the cone. What is it about the shape of the cone that could explain this?' Groups have 3 minutes to discuss and write a group hypothesis in one sentence.</p>                                                                                                                                                                                                                                                                                                                                                                                                               | <p>this question' — write on board: 'The cylinder holds EXACTLY three times as much as the cone. What is it about the shape that makes this happen?' Allow 3 minutes of group discussion. Circulate: listen for groups who mention 'the cone gets narrower', 'less space inside', or 'the cone is pointy at the bottom'. After 3 minutes, cold-call 3 groups to share their hypothesis sentence. After each, say: 'Interesting — hold that idea. We will test it mathematically in Lessons 2 and 3.' Do NOT resolve the question yet. Say: 'What you have just observed is the puzzle at the heart of this entire unit. Kericho tea farmers deal with this every single harvest.'</p>                                                                                                                                                                                                                           | <p>practice of constructing explanations from evidence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>the 3D interior shape will benefit from the net-building activity. Record group names against these three categories in your register margin for differentiation planning.</p>                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Comparing the Capacity and Volume of Three Containers (Metric Units)</a><br/>Source: CK-12<br/><a href="#">Search ARES</a><br/>for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Estimate volume (milliliters and liters)</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a><br/>for similar readings</p> | <p>The teacher facilitates a whole-class prerequisite knowledge review: the relationship between volume units and capacity units. Learners receive a mini reference card (or copy into their notebooks): <math>1 \text{ cm}^3 = 1 \text{ mL}</math>; <math>1\,000 \text{ cm}^3 = 1 \text{ L}</math>; <math>1 \text{ m}^3 = 1\,000 \text{ L}</math>. Using the actual capacity of the cylindrical drum (<math>\approx</math> 9.4 L), learners individually convert: (a) 9.4 L <math>\rightarrow</math> mL; (b) 9.4 L <math>\rightarrow</math> <math>\text{cm}^3</math>; (c) 3.1 L (cone) <math>\rightarrow</math> <math>\text{cm}^3</math>. They then compare: 'If Container A holds 9 400 <math>\text{cm}^3</math> and Container B holds 3 100 <math>\text{cm}^3</math>, what is the ratio A : B in simplest form?' Learners write this on their sheet. A volunteer comes to the board to show the working. The class checks. The teacher then poses: 'So if we know the volume formula for the cylinder, could we use this ratio to find the formula for the cone?' Learners write this as a question on their sheet — it is the bridge to Lesson 2.</p> | <p>Say: 'Before we can solve the Kericho farmers' puzzle mathematically, we need to make sure we all speak the same language of measurement. Let me show you the reference card.' Display or dictate the three conversion facts. Say: 'Using the actual capacities we observed, I want you to work these three conversions alone — silent work, 2 minutes.' After 2 minutes: 'Now the ratio — simplest form — 30 more seconds.' Cold-call one learner for (a), a different learner for (b), a third for (c), and a fourth for the ratio. For each, ask: 'Can someone tell me if they agree or disagree with that answer — and why?' Wait 10 seconds before accepting a response. After confirming 3 000 : 1 000 <math>\rightarrow</math> 3 : 1, say: 'This ratio — 3 to 1 — is not a coincidence. It is a mathematical law. Your job over the next few lessons is to discover where that law comes from and</p> | <p>Anchored prerequisite review: Rather than reviewing unit conversions in isolation, the teacher uses the actual data from the phenomenon as the numbers for the conversion exercises. This anchors prerequisite knowledge directly to the driving question context, ensuring learners see <math>\text{cm}^3</math> and L not as abstract units but as the language for describing the Kericho containers. The ratio calculation plants the seed of the core algebraic relationship before formal derivation.</p> | <p>Cold-call responses to the three conversions reveal: (a) learners who correctly convert all three are prerequisite-ready; (b) learners who confuse <math>\text{cm}^3</math> with cm or forget the factor of 1 000 need a brief individual follow-up — teacher writes the conversion ladder on a sticky note for those learners to keep in their notebook; (c) if more than 6 learners cannot state the simplest ratio, pause and reteach the conversion bridge using: 'How many mL are in 1 L? Good. So how many <math>\text{cm}^3</math> are in 9 400 mL?' before proceeding.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Volume of Cylinders (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each learner writes two questions on separate sticky notes (or torn pieces of paper if sticky notes are unavailable): one question about why the containers hold different amounts, and one question about how the shape affects the volume. Learners post their sticky notes on the class Driving Question Board — a large sheet of manila paper divided into three columns: 'WHAT I NOTICE', 'WHAT I WONDER', 'WHAT I NEED TO LEARN'. The teacher reads aloud five representative questions and, with the class, sorts them into the three columns. The teacher then formally posts the unit Driving Question in large print at the top of the board: 'Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?' The class reads the driving question aloud together once. Learners copy it into their notebooks and underline or circle any words they are unsure about.</p> | <p>Distribute two sticky notes (or paper squares) per learner. Say: 'On your first note, write a question that begins with WHY — something about these containers that you genuinely do not yet understand. On your second note, write a question that begins with HOW — something you would need to learn to design a better container for the Kericho farmers.' Allow 2 minutes of silent writing. Say: 'Come and post your notes — WHY questions on the left, HOW questions on the right. If you are not sure which column, post it in the middle.' Allow 2 minutes for posting. Read five questions aloud — choose ones that represent the range: one very specific ('Why is it exactly 3 times?'), one broad ('How do you calculate the volume of a weird shape?'), one applied ('How do farmers know which container to buy?'). After reading each, say: 'Which column do we think this belongs in?' Take a quick show of hands. After sorting, say: 'Every one of your questions is valid. Over the next lessons, we will answer each of them — and you will track your own growing understanding right here on this board.' Formally post and read the unit Driving Question. Say: 'This board will grow every lesson. By the last lesson of this unit, every sticky note on it will have an answer — an answer you worked out yourself.'</p> | <p>Public Question Mapping: The DQB makes learner thinking visible and collective. By sorting questions into NOTICE / WONDER / NEED TO LEARN, learners practise distinguishing observations from inferences and identifying knowledge gaps — core practices of mathematical inquiry. The board also serves as a living formative assessment tool: the teacher can read it after class to plan the sequence of lessons 2–7.</p> | <p>Scan sticky notes for question quality: (a) questions that name specific shapes ('Why does the cone get so much smaller?') show learners are attending to geometry — these learners are ready for formula derivation; (b) questions that are purely procedural ('What is the formula?') indicate learners need more phenomenon engagement — revisit the images at the start of Lesson 2; (c) absence of any HOW questions from a learner suggests they may not yet see mathematics as a design tool — check in with those learners during Lesson 2's group work.</p> |
| <b>Model Building</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | As a closing activity, learners individually draw a first-draft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Say: 'To close today, I want you to draw and label the three                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Initial Model Snapshot: Drawing a labelled model before instruction                                                                                                                                                                                                                                                                                                                                                            | Review initial model sentences after class. Categorise: (a) causal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



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| <p><b>Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a cone</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Volume of the Cone (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>annotated sketch of the three Kericho containers in their notebooks — a cylinder, a cone (apex down), and a frustum (bucket shape). They label: height, top diameter, and any other dimension they think might matter. Beneath the sketch they write one sentence that is their current best explanation for why the containers hold different amounts. This is their Initial Model — it will be revised in every subsequent lesson. Learners fold the page and write 'INITIAL MODEL — Lesson 1' on the outside so it can be found easily for future revision.</p> | <p>containers from memory — cylinder, cone, bucket — and write your current best explanation for the capacity difference. This is not a test — it is a snapshot of your thinking today, and you will improve it every lesson.' Allow 4 minutes of quiet drawing and writing. Circulate and read sentences without correcting — simply note them. After 4 minutes: 'Fold the page, write INITIAL MODEL — Lesson 1 on the outside, and leave it in your notebook. Next lesson, you will open it and see how your thinking has changed.' Close the lesson by pointing to the Driving Question Board: 'The Kericho farmers need a better container. You are the mathematicians who are going to design it. See you next lesson.'</p> | <p>forces learners to externalise their mental model of the containers and their current causal reasoning. Because this model will be physically revisited and revised in each lesson, it creates a tangible record of conceptual growth — a hallmark of model-based learning and a powerful motivator for learners who can see their own progress.</p> | <p>language ('The cone holds less because it narrows to a point so there is less space') — strong starting point; (b) comparative language only ('The cylinder is bigger than the cone') — learner is observing but not yet explaining; (c) no dimensional reasoning at all — learner will need additional scaffolding in Lesson 2 with a physical cone and cylinder. Use this to set the starting groups for Lesson 2's hands-on measurement activity.</p> |
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## D. TEACHER REFLECTION

After the lesson, consider the following: (1) How many learners predicted the correct ranking ( $A > C > B$ )? If fewer than one-third were correct, the phenomenon is sufficiently surprising — good. If most were correct, probe whether they already know the cone formula, as you may need to accelerate to Lesson 3 faster. (2) Were learners able to articulate WHY questions, or did their DQB notes only contain HOW questions? A board dominated by HOW suggests learners see this as a procedure unit — revisit the phenomenon image at the start of Lesson 2 and ask: 'What is it about the cone's shape — not its formula — that causes it to hold less?' (3) Review the initial model sentences: what is the modal level of explanation across the class? This tells you how much time to spend on the physical cylinder-cone water-pouring demonstration in Lesson 2. (4) Were any learners disengaged during the prediction phase? Consider assigning those learners the role of 'official recorder' on the DQB in Lesson 2 to increase buy-in. (5) Did the prerequisite review of  $\text{cm}^3 \leftrightarrow \text{L}$  reveal significant gaps? If more than 8 learners struggled with the conversions, begin Lesson 2 with a 5-minute targeted review using the Kericho container data before moving to formula derivation.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>What did I observe?</b>       | Three containers used by Kericho tea farmers — a cylindrical drum (A), a conical sisal cover (B), and a metal frustum bucket (C) — all with roughly the same height and opening diameter, were shown filled to the brim. Container A held $\approx 9.4 \text{ L}$ , Container B held $\approx 3.1 \text{ L}$ , and Container C held $\approx 6.5 \text{ L}$ . The cylinder held almost exactly three times as much as the cone.            |
| <b>What did I learn?</b>         | Volume is measured in $\text{cm}^3$ or $\text{m}^3$ ; capacity is measured in $\text{mL}$ or $\text{L}$ . The conversion relationships are: $1 \text{ cm}^3 = 1 \text{ mL}$ , $1\,000 \text{ cm}^3 = 1 \text{ L}$ , and $1 \text{ m}^3 = 1\,000 \text{ L}$ . The ratio of the cylinder's capacity to the cone's capacity (same height, same base) is $3 : 1$ — a fact that is not yet explained but will be investigated across this unit. |
| <b>How does this explain the</b> | We cannot yet fully explain why containers of nearly the same height and opening diameter hold such different amounts. Our                                                                                                                                                                                                                                                                                                                 |

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| <b>phenomenon?</b> | current best explanation is that the cone narrows to a point, leaving less interior space, while the cylinder stays the same width all the way down. We have posted our questions on the Driving Question Board and will build toward a complete mathematical explanation in the lessons ahead. |
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## LESSON 2 (80 minutes (2 × 40-minute lessons)): Cone vs. Cylinder — Deriving the Cone Volume Formula

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To establish, through direct observation and guided reasoning, that the volume of a cone is exactly one-third the volume of a cylinder sharing the same base radius and height, and to apply the derived formula $V = \frac{1}{3}\pi r^2 h$ to calculate the capacity of the Kericho conical tea-cover.                                                                                                                                                                                                                                                                                                                              |
| <b>Knowledge</b>            | Learners will know that: (a) a cone and a cylinder of equal base radius and height satisfy $V(\text{cylinder}) = 3 \times V(\text{cone})$ ; (b) the volume formula for a cone is $V = \frac{1}{3}\pi r^2 h$ ; (c) capacity in litres is obtained by converting $\text{cm}^3$ using 1 litre = 1 000 $\text{cm}^3$ .                                                                                                                                                                                                                                                                                                                   |
| <b>Skills</b>               | Learners will be able to: (a) measure the base radius and height of a cone and cylinder; (b) use repeated pouring to gather quantitative evidence of the 1:3 ratio; (c) derive and write the cone volume formula algebraically; (d) substitute given dimensions into $V = \frac{1}{3}\pi r^2 h$ and convert the result to litres; (e) update the class Driving Question Board with new evidence.                                                                                                                                                                                                                                     |
| <b>Attitudes</b>            | Learners will: appreciate that careful measurement and systematic observation lead to reliable mathematical generalisations; value the relevance of geometry to the daily work of Kenyan tea farmers; demonstrate patience, precision and collaborative respect during the pouring demonstration.                                                                                                                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Purpose in Storyline</b> | In Lesson 1 learners recorded their initial predictions about why the three Kericho containers hold different amounts despite looking similar. Lesson 2 is the first major evidence-gathering lesson: learners directly observe the cone–cylinder pouring experiment, derive $V = \frac{1}{3}\pi r^2 h$ , and use it to calculate the actual capacity of the conical sisal cover. This gives them the first precise number to anchor the rest of the unit and deepens the puzzle — the cone holds only one-third as much as the cylinder, so why does the metal bucket (frustum) hold more than the cone but less than the cylinder? |
| <b>Safety Notes</b>         | Use plastic or metal containers for the pouring demonstration to avoid glass breakage. Wipe up any water spills immediately to prevent slipping. If food colouring is added to the water for visibility, remind learners it may stain clothing.                                                                                                                                                                                                                                                                                                                                                                                      |

### B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 1 predictions recorded on the Driving Question Board about the three Kericho tea-farmer containers. The teacher introduces a physical demonstration: a transparent plastic cylinder and a matching cone (identical base radius  $r$  and height  $h$ ). Learners predict how many cones of water will fill the cylinder, then observe the repeated-pouring experiment three times. Using the known cylinder formula  $V = \pi r^2 h$  as an anchor, learners deduce that  $V(\text{cone}) = \frac{1}{3}\pi r^2 h$ . They then apply the formula to the specific dimensions of the Kericho conical sisal cover ( $r = 21$  cm,  $h = 42$  cm), calculate its volume in  $\text{cm}^3$ , convert to litres, and record the result as the first confirmed data point on the Driving Question Board. The lesson closes with a model-building update and targeted formative exit questions.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                                | Teacher Moves                                           | Sensemaking Strategy                                            | Formative Assessment Strategy                                         |
|----------------------|-------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Predict Phase</b> | Learners individually write a prediction in their exercise books: | Display the two physical containers (cone and cylinder, | Think-Pair-Share with class tally: activates prior knowledge of | Observable indicator: learner can correctly label $r$ and $h$ on both |

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| <p> <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume of cones</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>'How many full cones of water do you think it will take to completely fill the matching cylinder?' They then share predictions with a partner (Think-Pair-Share) and the class records a show-of-hands tally on the board (options: 1, 2, 3, 4, more than 4). Learners also sketch the cone and cylinder side by side, labelling what they already know: same opening diameter, same height — reinforcing the connection to the Kericho phenomenon.</p>                                                                                                                                                                                                                                                                                                                                                              | <p>same <math>r</math> and <math>h</math>) prominently. Say: 'These represent the tall cylindrical drum and the conical sisal cover the Kericho farmers use. They have exactly the same opening width and the same height. Before we pour a single drop, I want your best mathematical guess — how many cones of water will fill the cylinder?' Allow 10 seconds of silent thinking (WAIT TIME). Cold-call 4 learners for their predictions and record each on the board. Then say: 'Hold that prediction. We are going to find out together right now.'</p>                                                                                                                                                                                                                                                                                              | <p>cylinder volume (<math>V = \pi r^2 h</math> from Junior School) and creates cognitive dissonance when the result surprises learners who predicted 1 or 2. The public tally makes predictions visible and personally invested.</p>                                                                                                                                                                     | <p>shapes in their sketch. Listen for whether learners spontaneously mention <math>\pi r^2 h</math> — this reveals recall of the cylinder formula, which will be needed in the Explain phase.</p>                                                                                                                                                                                                                                     |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Volume of a cone</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Volume of Cylinders (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>     | <p>The teacher (or a volunteer learner) fills the cone to the brim with water (coloured blue with food dye for visibility) and pours it into the cylinder. This is repeated a second and third time. Learners complete an observation table in their books with three columns: Pour Number   Water level in cylinder (estimate as fraction of height)   Running total (cones poured). After the third pour the cylinder is exactly full. Learners record the key observation: 3 cones = 1 cylinder. A short 3-minute silent reading extract (one paragraph) is then provided, describing how Kenyan agricultural extension officers at the Kericho Tea Development Authority use volume calculations to specify container sizes for tea-leaf collection — grounding the mathematics in a real professional context.</p> | <p>Before pouring, say: 'Watch the water level carefully after each pour. In your table, write a fraction — is it one-quarter full, one-half full, two-thirds full? Be precise.' Carry out Pour 1. WAIT 10 seconds after the water settles. Ask: 'What fraction of the cylinder is now filled? Turn to your neighbour — 10 seconds.' Cold-call 3 learners. Carry out Pour 2 and repeat questioning. Carry out Pour 3 — the cylinder is full. Say: 'What did we observe?' Cold-call 2 learners and require them to state it as a complete sentence: 'Three cones of water exactly fill one cylinder of the same base and height.' Distribute the short reading extract on the Tea Development Authority. After 3 minutes say: 'Underline one sentence that connects what we just saw to why a farmer or an extension officer would need to know this.'</p> | <p>Structured observation table with fractional estimation: this forces quantitative attention at each stage rather than a single end-point observation, building the proportional reasoning needed to generalise the 1:3 ratio. The reading extract (Evidence from Text strategy) anchors the abstract observation in a professional Kenyan agricultural context, reinforcing the driving question.</p> | <p>Observable indicator: learner's observation table shows entries of approximately <math>\frac{1}{3}</math> full after Pour 1, <math>\frac{2}{3}</math> full after Pour 2, and completely full after Pour 3. Listen for learners who state the ratio as '3 cones = 1 cylinder' unprompted — they are ready for the algebraic derivation. Flag learners who write '2 cones' or express surprise — they need the derive step most.</p> |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a><br/> Source: Khan</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Learners work through a guided derivation in three steps written in their books: (Step 1) Write the cylinder formula: <math>V(\text{cylinder}) = \pi r^2 h</math>. (Step 2) Use the observation:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Write on the board: 'We know <math>V(\text{cylinder}) = \pi r^2 h</math>. We just saw <math>V(\text{cylinder}) = 3 \times V(\text{cone})</math>. So <math>V(\text{cone}) = ?</math>' WAIT 15 seconds of silent working time. Cold-call 2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Guided algebraic derivation (Claim-Evidence-Reasoning): learners construct the formula themselves using the observation as evidence rather than receiving it</p>                                                                                                                                                                                                                                      | <p>Observable indicator: learner's written derivation shows all three steps with correct algebraic manipulation. Correct numerical answer: <math>V \approx 19\,373\text{ cm}^3 \approx 19.4</math></p>                                                                                                                                                                                                                                |

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| <p>Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume of Cylinders Practice</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                  | <p><math>V(\text{cylinder}) = 3 \times V(\text{cone})</math>. (Step 3) Therefore: <math>V(\text{cone}) = \pi r^2 h \div 3 = \frac{1}{3} \pi r^2 h</math>. Learners then immediately apply the formula to the Kericho conical sisal cover: given <math>r = 21</math> cm and height <math>h = 42</math> cm, calculate (a) volume in <math>\text{cm}^3</math> and (b) capacity in litres. They compare this figure to the cylindrical drum (same <math>r = 21</math> cm, <math>h = 42</math> cm) calculated in a worked example on the board, and record the ratio 1:3 numerically using actual values. Learners also write one sentence explaining to a Kericho farmer why the cone-shaped cover cannot be used to store the same amount of water as the cylindrical drum even though both containers 'look the same size'.</p> | <p>learners to complete the algebra on the board. Confirm the formula: <math>V = \frac{1}{3} \pi r^2 h</math>. Then say: 'Now let us use this formula for the actual Kericho conical sisal cover. I'll give you the measurements: radius 21 cm, height 42 cm. Work it out — you have 5 minutes.' Circulate and look at 6 books. After 5 minutes, cold-call one learner to write their working on the board. Guide the class to check: <math>V = \frac{1}{3} \times \pi \times 21^2 \times 42 = \frac{1}{3} \times \pi \times 441 \times 42 \approx \frac{1}{3} \times 58120 \approx 19373 \text{ cm}^3 \approx 19.4</math> litres. Ask: 'So the conical cover holds about how many litres?' Cold-call 3 learners. Then say: 'Write one sentence — pretend you are explaining this to a Kericho tea farmer who has never studied mathematics. Why does the cone hold less?' WAIT 10 seconds, then pair-share before cold-calling 2 learners to read their sentences aloud.</p> | <p>as a given rule. This strategy builds understanding of why the formula has a factor of <math>\frac{1}{3}</math>, not just what the formula is. The farmer-explanation sentence forces learners to translate formal mathematics into contextual meaning, deepening conceptual ownership.</p>                                                                             | <p>litres. Listen for learners who write '<math>V = \pi r^2 h \div 3</math>' without connecting it to the observation — prompt them: 'What did the pouring tell us first?' Flag learners who forget to divide by 3 or use diameter instead of radius — these are the two most common errors at this stage.</p>                                                                                                                                |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/> <a href="#">Volume of a cone</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Volume of the Cone (PLIX)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>The class DQB (a large chart on the wall) has three columns: What we observed   What we now know   What we still wonder. Each learner writes two sticky notes: one for 'What we now know' (e.g. 'The conical cover holds about 19.4 litres — exactly <math>\frac{1}{3}</math> of the cylinder') and one for 'What we still wonder' (e.g. 'The metal bucket holds more than the cone — is there a formula for that shape too?'). Sticky notes are placed on the DQB. The teacher reads two or three aloud and connects them explicitly to the driving question about the frustum-shaped bucket.</p>                                                                                                                                                                                                                         | <p>Point to the original driving question displayed above the DQB: 'Why do the three Kericho containers look nearly the same size but hold such different amounts?' Say: 'We have just answered part of this question for the cone. Write your two sticky notes — one for what you now know for certain, and one question you still have. You have 3 minutes.' After placing notes, read 2 from 'What we now know' and 2 from 'What we still wonder' aloud. Say: 'Notice that many of you are asking about the metal bucket — that shape between the cone and the cylinder. That is exactly where we are going next. Keep that question burning.'</p>                                                                                                                                                                                                                                                                                                                         | <p>Driving Question Board update (public knowledge-tracking): making individual understanding visible to the whole class builds collective sense of progress, validates students' remaining confusion as productive, and creates explicit anticipation for the frustum lessons. The act of writing forces metacognitive reflection on what changed during this lesson.</p> | <p>Observable indicator: learner's 'What we now know' sticky note includes a specific number (litres or <math>\text{cm}^3</math>) or the formula <math>V = \frac{1}{3} \pi r^2 h</math> — not just a vague statement. A 'What we still wonder' note that references the bucket/frustum indicates readiness for Lesson 3. Collect 5–6 sticky notes from learners who said least during the lesson to gauge silent learners' understanding.</p> |
| <p><b>Model</b></p>                                                                                                                                                                                                                                                                                                                                               | <p>Learners retrieve the cumulative</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Say: 'Get out the diagram you</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Cumulative annotated model</p>                                                                                                                                                                                                                                                                                                                                          | <p>Observable indicator: the diagram</p>                                                                                                                                                                                                                                                                                                                                                                                                      |

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| <p><b>Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Volume of Cones (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>diagram they started in Lesson 1 (a side-by-side sketch of the three Kericho containers: cylinder, cone, and bucket). Today they add three pieces of information to the diagram: (a) the formula <math>V = \frac{1}{3}\pi r^2 h</math> written inside or beside the cone, (b) the calculated capacity <math>\approx 19.4</math> L' labelled on the conical cover, and (c) a double-headed arrow between the cone and the cylinder labelled 'x3' to show the ratio. Learners also add a question-mark annotation on the bucket: 'V = ? (more than cone, less than cylinder)'. This partial model will be updated in every subsequent lesson until the frustum formula is derived in Lesson 4.</p> | <p>started in Lesson 1. Today you are going to make it more precise — add the formula, the calculated number, and the ratio arrow. You have 4 minutes.' Circulate and check that the x3 arrow is placed correctly (from cone to cylinder, not the reverse). Ask one learner: 'If I drew another arrow pointing from the cylinder back to the cone, what fraction would I write on it?' (Expected answer: <math>\frac{1}{3}</math>.) After 4 minutes, display a model diagram on the board and say: 'Your diagram should look something like this. The question mark on the bucket is the most important thing on your whole page — that is the mystery we are solving together over the next four lessons.'</p> | <p>(Visualising Mathematical Relationships): progressively adding quantitative and formulaic information to a single persistent diagram builds a coherent mental model of how the three shapes relate. The deliberate question-mark annotation on the frustum maintains productive inquiry tension and connects every lesson to the overarching driving question.</p> | <p>shows the correct formula <math>V = \frac{1}{3}\pi r^2 h</math> on the cone, a numerically correct capacity label (within rounding), and the x3 ratio arrow pointing from cone to cylinder. A question-mark annotation on the bucket indicates the learner understands the frustum is the remaining puzzle. Collect 4 diagrams at the end of the lesson as a formative artefact to review before Lesson 3.</p> |
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## D. TEACHER REFLECTION

After the lesson, consider: (1) During the pouring demonstration, did all learners have a clear line of sight to the water level, or should the cylinder be placed on a raised surface next time? (2) How many learners predicted '3' before the demonstration — and did those who predicted otherwise show genuine surprise? Surprise is a sign of authentic inquiry; if most learners already knew the answer, consider whether this group is ready to move to pyramids faster. (3) Were learners able to distinguish between radius and diameter when substituting into  $V = \frac{1}{3}\pi r^2 h$ ? Errors here are diagnostic of a gap from Junior School work on circles that will compound in Lessons 3–5. (4) Review the sticky notes from the DQB: do learners' 'What we still wonder' questions specifically mention the bucket/frustum shape, or are they still focused on the cone–cylinder comparison? If fewer than half mention the frustum, spend 5 minutes at the start of Lesson 3 re-reading the driving question aloud and pointing to the bucket explicitly. (5) Did the real-world connection to the Kericho Tea Development Authority feel genuine to learners, or did it feel artificially inserted? Ask two learners informally after class.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <p><b>What did I observe?</b></p>                   | <p>Learners filled a cone with water and poured it into a matching cylinder three times, directly observing that the cylinder was exactly full after the third pour — demonstrating <math>V(\text{cylinder}) = 3 \times V(\text{cone})</math> using containers that represent the Kericho tea-farmer drum and conical sisal cover.</p>                                                              |
| <p><b>What did I learn?</b></p>                     | <p>Learners derived the cone volume formula <math>V = \frac{1}{3}\pi r^2 h</math> algebraically from the observed 1:3 ratio, calculated the capacity of the Kericho conical cover (<math>r = 21</math> cm, <math>h = 42</math> cm) as approximately <math>19\,373\text{ cm}^3 \approx 19.4</math> litres, and confirmed it holds exactly one-third the volume of the matching cylindrical drum.</p> |
| <p><b>How does this explain the phenomenon?</b></p> | <p>Learners explained in writing why the conical sisal cover holds far less water than the cylindrical drum despite identical height and opening width, and updated their cumulative container diagram with the formula, the calculated capacity, and a x3 ratio arrow —</p>                                                                                                                        |



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|  | leaving a deliberate question mark on the metal bucket (frustum) as the next mathematical puzzle to solve. |
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## LESSON 3 (80 minutes): Volume of Pyramids: Unlocking the One-Third Relationship

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Purpose</b>              | Learners establish and apply the formula $V = \frac{1}{3} \times \text{base area} \times \text{height}$ for square-based pyramids by physically comparing pyramid and cuboid models, then use the formula to compute volumes and capacities of pyramidal grain stores — building the mathematical bridge between the cone-cylinder relationship (Lesson 1) and the frustum (Lessons 5–7).                                                                                                                                                                              |
| <b>Knowledge</b>            | Learners determine the volume and capacity of pyramids using the formula $V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}$ , and convert between volume units ( $\text{cm}^3$ , $\text{m}^3$ ) and capacity units (litres, millilitres).                                                                                                                                                                                                                                                                                                  |
| <b>Skills</b>               | Learners measure dimensions of pyramid and cuboid models, substitute values into the volume formula, perform unit conversions, and compare computed capacities with Lesson 1 predictions recorded on the Driving Question Board.                                                                                                                                                                                                                                                                                                                                       |
| <b>Attitudes</b>            | Learners appreciate that the mathematical relationship between a pyramid and a cuboid mirrors the cone-cylinder relationship, recognising that elegant patterns in geometry have practical value for Kenyan farmers designing storage containers.                                                                                                                                                                                                                                                                                                                      |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Purpose in Storyline</b> | In Lesson 1, learners predicted why the cylindrical drum holds three times the water of the conical cover. In Lesson 2, they confirmed $V(\text{cone}) = \frac{1}{3} \times V(\text{cylinder})$ . Now in Lesson 3, they discover that the exact same one-third relationship exists between a square-based pyramid and its surrounding cuboid — deepening the pattern and preparing them to later 'slice' a frustum from a pyramid (Lesson 5). This lesson also introduces grain-store pyramids as a second Kenyan context alongside the Kericho tea-farmer containers. |
| <b>Safety Notes</b>         | Learners handle rulers, scissors and card-stock when constructing or measuring models. Remind learners to pass scissors handle-first and to keep cutting surfaces flat on the desk. Water used in any volume-by-displacement demonstration should be in a stable, low-sided tray to avoid spillage on exercise books.                                                                                                                                                                                                                                                  |

### B. LESSON OVERVIEW

Learners open with a quick written prediction connecting today's pyramid to the cone-cylinder pattern from Lesson 2. They then gather evidence using a hands-on pair activity: a card-stock square-based pyramid and a cuboid of identical base and height are filled with sand/rice to prove  $V(\text{pyramid}) = \frac{1}{3} \times V(\text{cuboid}) = \frac{1}{3} \times \text{base area} \times \text{height}$ . In the Explain phase, learners apply the formula to three pyramidal grain-store problems set in the Rift Valley, computing volumes in  $\text{cm}^3/\text{m}^3$  and converting to litres. They compare their computed capacities with their Lesson 1 DQB predictions, annotate errors, and close by adding two new 'How do we know...?' questions to the board. The lesson ends with a Model Building entry in which learners revise their Lesson 2 proportional diagram to include the pyramid-cuboid trio alongside the cone-cylinder trio.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                  | Learner Experience                                                                          | Teacher Moves                                                                                         | Sensemaking Strategy                                                                                       | Formative Assessment Strategy                                                                  |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Volume of</a> | Learners receive a half-page prompt card showing a sketch of a square-based pyramid and its | Distribute prompt cards face-down. Say: 'Turn your card over. You have 3 minutes — silent, individual | Prediction with Justified Reasoning — learners are required to anchor their numeric prediction to a causal | Observable indicator: learner has written a numeric prediction (fraction or litre value) AND a |

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| <p><a href="#">pyramids intuition</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Surface Area and Volume of Pyramids Practice</a><br/>Source: CK-12<br/><a href="#">Search ARES</a> for similar readings</p>                  | <p>enclosing cuboid (same base dimensions and height), both labelled with base side <math>s = 12</math> cm and height <math>h = 15</math> cm. Working individually and silently for 3 minutes, they write two predictions in their exercise books: (a) 'What fraction of the cuboid's volume do you think the pyramid holds? Give a reason linked to what we learned about cones and cylinders.' (b) 'If the cuboid holds 120 litres of grain, predict how many litres the pyramid holds.' After 3 minutes, learners share predictions with a shoulder partner for 2 minutes, then the teacher cold-calls 4 learners to share with the class.</p>                                                                                                                                                                                                                                                                       | <p>thinking. Write your prediction AND your reason. I am looking for a connection to what we proved last lesson about cones and cylinders.' Start a visible 3-minute timer. Circulate and read over shoulders; note 2 learners who wrote 'one-third' and 2 who wrote a different fraction — you will cold-call all four. After 3 minutes say: 'Share with your shoulder partner for 2 minutes — convince each other.' After partner talk, say: 'I am going to hear from four people. Amina — what did you predict and why? [WAIT 12 seconds.] Thank you. James — do you agree or disagree with Amina? [WAIT 10 seconds.]' Cold-call 4 learners total. Record predictions in two columns on the board: 'One-third camp' and 'Other fraction camp'. Do NOT confirm or deny. Say: 'Let us gather evidence to settle this.'</p> | <p>reason (the cone-cylinder pattern), activating prior knowledge and creating a cognitive need to test the claim. Recording two 'camps' on the board makes the disagreement visible and motivates the evidence phase.</p>                                                                                                                                          | <p>written reason that references either 'one-third', 'cone', or 'cylinder'. Teacher scans exercise books during circulation; learners who have only a number with no reason are prompted: 'Add one sentence — why do you think that?'</p>                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Volume of a Pyramid - Overview</a><br/>Source: CK-12<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/><a href="#">Surface Area and Volume of Pyramids (PLIX)</a><br/>Source: CK-12<br/><a href="#">Search ARES</a> for similar readings</p> | <p>Pairs receive one pre-made open-top card-stock square-based pyramid and one open-top cuboid of identical base (<math>10\text{ cm} \times 10\text{ cm}</math>) and height (<math>12\text{ cm}</math>), plus a cup of dry rice/sand and a 250 ml measuring jug. Step 1 (5 min): Pairs fill the pyramid level-full with rice, then pour it into the cuboid. They repeat and tally how many full pyramids of rice it takes to fill the cuboid completely. Step 2 (5 min): Pairs record their tally, compute the cuboid volume (<math>V = l \times w \times h = 10 \times 10 \times 12 = 1\,200\text{ cm}^3</math>) and use their tally to write the fraction pyramid/cuboid. Step 3 (5 min): Pairs measure their pyramid's base side and height with a ruler, then use <math>V = \frac{1}{3} \times s^2 \times h</math> to compute the theoretical pyramid volume and compare with the experimental fraction. Step 4</p> | <p>Before distributing materials say: 'You are going to do what scientists do — let the physical world tell us the answer. Your job is to fill carefully and count carefully. No spilling.' Distribute kits. After 2 minutes of filling say: 'Pause. Look at your tally so far. Whisper to your partner: is the pyramid going to need more or fewer fills than you predicted?' [WAIT 10 seconds.] Continue circulating. When most pairs have completed 3 fills say: 'If you have filled three times and the cuboid is not yet full, what does that tell you? [WAIT 12 seconds.] Keep going.' After Step 2, pause the class: 'Hands up — who got exactly 3 fills?' [Count hands aloud.] 'Who got close to 3 but not exactly — what might cause that?' [Cold-call 2 learners, WAIT 10</p>                                     | <p>Hands-On Physical Modelling with Formula Verification — the concrete rice-filling activity generates experiential data (3 fills = one-third), which learners then reconcile with the algebraic formula. The two-step approach (experimental → formula) builds conceptual understanding before procedural fluency, preventing formula-as-magic-rule thinking.</p> | <p>Observable indicators: (1) Learner's Evidence Table shows a tally of 3 (<math>\pm 0.2</math> tolerance noted) and the fraction <math>\frac{1}{3}</math> written explicitly. (2) Learner has correctly computed <math>V(\text{cuboid}) = 1\,200\text{ cm}^3</math> and <math>V(\text{pyramid}) = 400\text{ cm}^3</math> from the formula. (3) Grain-store card A: learner has written <math>V = \frac{1}{3} \times 9 \times 2.4 = 7.2\text{ m}^3 = 7\,200\text{ litres}</math>. Teacher checks Step 4 answers by circulating and initialling correct work before the class moves on.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (5 min): Each pair reads 'Grain-store card A' — a diagram of a pyramidal grain store used in Eldoret with base 3 m × 3 m and height 2.4 m — and uses the formula to calculate the volume in m <sup>3</sup> and convert to litres (1 m <sup>3</sup> = 1 000 litres). Pairs record all steps in a structured 'Evidence Table' on a printed worksheet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | seconds each.] After Step 3 say: 'Compare your experimental fraction with your formula fraction. Are they the same? If not, write one sentence explaining why.' After Step 4 say: 'Now connect this to our Kericho farmers. The tea-farmer uses a conical cover and the Eldoret grain-store is a pyramid. What do these two shapes have in common mathematically?' [Cold-call 2 learners, WAIT 12 seconds each.]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of pyramids intuition</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume of prisms and pyramids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners work in groups of four on three tiered application problems, all set in Kenyan contexts.</p> <p>Problem 1 (Accessible): A pyramidal heap-cover used by tea pickers in Kericho has a square base of side 80 cm and a height of 60 cm. Calculate its volume in cm<sup>3</sup> and convert to litres. Problem 2 (Standard): A grain store in Kisumu is built in the shape of a square pyramid with base 4 m × 4 m and height 3 m. The farmer wants to store 20 m<sup>3</sup> of maize. How many such stores does she need? Problem 3 (Extended): A community in the Rift Valley uses two pyramidal stores: Store A has base 2.5 m × 2.5 m and height 3.6 m; Store B has base 3 m × 3 m and height 2 m. Which store holds more githeri (dry beans and maize)? By how many litres? Each group attempts all three, then one group presents Problem 2 and another presents Problem 3 to the class. Presenters must say: 'Our formula is...', 'We substituted...', and 'Our answer means that in the context of the farmer...'</p> | <p>Say: 'These three problems are all about Kenyan farmers storing food. Every calculation you do must end with a sentence that means something to a real farmer — not just a number.' Set a 12-minute group work timer. Circulate to Problem 2 groups first (most common error: forgetting to divide by 3). Prompt: 'You wrote <math>V = 4 \times 4 \times 3</math>. Is there a step missing? Look at your Evidence Table.' [WAIT 12 seconds.] For Problem 3 prompt struggling groups: 'Calculate each store separately first. Then subtract.' After 12 minutes select two groups for presentations. After each presentation ask the class: 'Do you agree with their formula substitution? Show me thumbs up, sideways or down.' [Pause 8 seconds.] Cold-call one 'sideways' learner: 'What concerns you?' [WAIT 12 seconds.] After both presentations ask the whole class: 'Look at the cone in our Kericho phenomenon and the pyramid in these problems. What is the shared mathematical secret?' [WAIT 15 seconds.] Cold-call 3 learners. Consolidate on the board: <math>V(\text{pyramid}) = \frac{1}{3} \times b^2 \times h</math> alongside <math>V(\text{cone}) = \frac{1}{3} \times \pi r^2 \times h</math>,</p> | <p>Structured Group Problem Solving with Public Presentation — tiered problems ensure all learners access the formula while the presentation requirement ('our answer means that...') forces contextualisation. Displaying <math>V(\text{pyramid})</math> and <math>V(\text{cone})</math> side-by-side on the board makes the structural pattern explicit, supporting transfer to frustums in later lessons.</p> | <p>Observable indicators: (1) Problem 1: <math>V = \frac{1}{3} \times 80^2 \times 60 = 128\,000\text{ cm}^3 = 128\text{ litres}</math> — learner has shown the division-by-3 step explicitly. (2) Problem 2: <math>V = \frac{1}{3} \times 16 \times 3 = 16\text{ m}^3</math>; <math>20 \div 16 = 1.25</math>, so 2 stores needed — learner has interpreted the non-integer answer correctly in context. (3) Problem 3: Store A = 7 500 litres, Store B = 6 000 litres; A holds 1 500 litres more — learner has converted both to litres before comparing. Teacher stamps or ticks at least Problem 1 in each learner's book before the DQB phase.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a Pyramid - Overview</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Volume of Pyramids: Fluctuating Height (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>The teacher uncovers the class DQB (begun in Lesson 1). Learners spend 3 minutes silently re-reading all current sticky notes. They then open their Lesson 1 prediction notebooks and compare their original guesses about the Kericho containers with what they now know from Lessons 1–3. Each learner writes on two sticky notes: (i) One 'We now know...' statement — something today's lesson confirmed or corrected from their Lesson 1 prediction. (ii) One 'We still need to figure out...' question — a gap that today's evidence did not yet answer (e.g., 'We know cone = <math>\frac{1}{3}</math> cylinder and pyramid = <math>\frac{1}{3}</math> cuboid, but how does the metal bucket — a frustum — fit this pattern?'). Learners post notes in the appropriate DQB columns. The teacher reads 3–4 new questions aloud and asks the class to vote (hands) on which question they most want to answer next.</p> | <p>Uncover the DQB. Say: 'Take 3 minutes — read every note on this board. Think about how today's lesson changes or confirms what is already here.' [Set 3-minute timer, enforce silence.] After 3 minutes: 'Open your Lesson 1 prediction page. What did you predict then? What do you know now? Write your two sticky notes.' [Allow 4 minutes.] Circulate and prompt learners whose 'We still need to figure out...' note is too vague (e.g., 'more about frustums'): 'Can you make it more specific — what exactly about the frustum puzzles you?' After posting, read 4 selected notes aloud. For each: 'Who agrees this is our next big question? [Hands.]' Highlight the winning question with a star sticker. Say: 'That starred question is exactly where Lessons 5, 6 and 7 will take us. Keep it in mind.'</p> | <p>Predict–Confirm–Question Cycle on the DQB — revisiting Lesson 1 predictions against Lesson 3 evidence makes learning progression visible to learners. The 'We still need to figure out...' notes sustain epistemic curiosity and give learners authorship over the unit's investigative direction, increasing engagement in subsequent lessons.</p>                                  | <p>Observable indicators: (1) Learner's 'We now know...' note references a specific numerical relationship (e.g., 'one-third', '<math>\frac{1}{3} \times</math> base area <math>\times</math> height') rather than a vague statement. (2) Learner's 'We still need to figure out...' note names the frustum or the metal bucket from the phenomenon. Teacher photographs the updated DQB at end of lesson for the class digital portfolio.</p>                                                                                        |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Pyramids Practice</a></p>                                                                             | <p>Learners return to the proportional diagram they built in Lesson 2 (a cylinder and cone drawn to scale showing the 3:1 volume ratio). They now extend that diagram by drawing a cuboid and its enclosed pyramid (same proportions) directly beside the cone-cylinder pair, labelling both with the <math>\frac{1}{3}</math> relationship. Below both pairs they write: 'Pattern: Pointed shape = <math>\frac{1}{3} \times</math> surrounding box, whether the base is a circle (cone) or a square (pyramid).' They then annotate the Kericho phenomenon image (printed on the worksheet) by drawing a dashed arrow from the metal bucket (frustum) with the label: 'NEXT: frustum = pyramid –</p>                                                                                                                                                                                                                            | <p>Distribute Lesson 2 diagrams (returned or re-drawn from notes). Say: 'Your model so far shows cones and cylinders. Today we add a new pair to the family — pyramid and cuboid. Draw them beside your existing pair and label the <math>\frac{1}{3}</math> relationship.' [Allow 6 minutes.] Circulate; prompt: 'Make sure your pyramid and cuboid are drawn to the same scale ratio as your cone and cylinder — does your drawing show the pyramid is clearly smaller than the cuboid?' After 6 minutes: 'Now look at the metal bucket — the frustum — in our Kericho picture. Add that dashed arrow. What question does your arrow ask?' [Allow 3 minutes.]</p>                                                                                                                                                       | <p>Cumulative Annotated Model Revision — learners physically extend a growing visual model across lessons, making cross-lesson knowledge integration tangible. The dashed arrow pointing to the frustum creates a forward-looking knowledge gap that primes learners for Lessons 5–7, reinforcing the unit's narrative coherence and the anchoring phenomenon's unresolved element.</p> | <p>Observable indicators: (1) Learner's model shows both pairs (cone-cylinder AND pyramid-cuboid) with the <math>\frac{1}{3}</math> label on both. (2) The written pattern statement uses the words 'pointed shape', '<math>\frac{1}{3}</math>', and either 'circle' or 'square' as the base descriptor. (3) The dashed arrow to the frustum is present and labelled with a question (not just 'frustum'). Teacher collects models at the end of the lesson to review overnight and return with written feedback before Lesson 4.</p> |

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| Source: CK-12<br><a href="#">Search ARES for similar readings</a> | smaller pyramid? Volume still to be determined.' Learners share their revised model with a partner for 2 minutes, giving one specific piece of praise and one question. | Pair share for 2 minutes: 'One praise, one question — be specific.' Cold-call 2 learners to share their partner's question: 'What question did your partner write on their arrow? [WAIT 10 seconds.]' Close: 'Your model is growing. Every lesson we add one more piece. By Lesson 7 your model will explain exactly why the metal bucket holds more than the cone but less than the cylinder — and you will be able to calculate exactly how much more.' |  |  |
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## D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the rice-filling activity, did all pairs obtain close to 3 fills? If several groups got 2 or 4 fills, what procedural issues (overfilling, poorly constructed models) should be addressed before Lesson 5? (2) In the Explain phase, did learners consistently write the  $\div 3$  step explicitly, or did many jump from  $V = b^2 \times h$  to the answer? If the latter, plan a short starter in Lesson 4 where learners must annotate a worked example showing every step. (3) Looking at the DQB notes: did learners' 'We still need to figure out...' questions point toward the frustum, or are there unexpected gaps (e.g., confusion about volume vs capacity units) that need to be addressed before moving to Lesson 4? (4) Which learners have not yet connected today's pyramid result back to the Kericho metal bucket? Identify them by name for targeted questioning in the next lesson's Predict Phase. (5) Were the three application problems appropriately tiered? If most groups did not reach Problem 3, consider whether Problem 2 needs scaffolding (e.g., a partially completed formula) in future iterations.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>What did I observe?</b>                   | Learners filled a card-stock square-based pyramid with rice and poured it into an identical-base-and-height cuboid, repeating until the cuboid was full. They recorded that it took 3 full pyramids to fill 1 cuboid, and used a ruler to measure the base and height of their pyramid model.                                                                                                                                                                                                                                                                                                |
| <b>What did I learn?</b>                     | $V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}$ ; the same one-third relationship that links a cone to its cylinder also links a square-based pyramid to its cuboid. Learners also practised converting pyramid volumes from $\text{cm}^3$ and $\text{m}^3$ into litres ( $1 \text{ m}^3 = 1\,000 \text{ litres}$ ) using grain-store problems from Eldoret, Kisumu, and the Rift Valley.                                                                                                                                                                     |
| <b>How does this explain the phenomenon?</b> | The one-third factor in both $V(\text{cone}) = \frac{1}{3} \times \pi r^2 \times h$ and $V(\text{pyramid}) = \frac{1}{3} \times b^2 \times h$ reflects a geometric pattern: any pointed solid (apex tapering to a point) holds exactly one-third of its surrounding box, regardless of whether the base is circular or square. This explains why the pyramidal grain stores used by Kenyan farmers hold far less grain than a similarly sized cuboid store — and raises the question of how to compute the volume of the frustum-shaped metal bucket, which lies between these two extremes. |



## LESSON 4 (80 minutes): Introducing Frustums of Cones: Deriving the Volume Formula and Solving the Bucket Mystery

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Purpose</b>              | Learners derive the volume formula for a frustum of a cone by reasoning from the cone-cylinder relationship established in earlier lessons, then apply it to calculate the exact capacity of the Kericho tea farmers' metal bucket — resolving the anchoring phenomenon.                                                                                                                                                                                                                                                                                                 |
| <b>Knowledge</b>            | Learners should be able to: (a) determine the volume and capacity of a frustum of a cone; (b) derive $V(\text{frustum}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ by subtracting the small cone from the large cone; (c) apply volume and capacity of frustums in real-life situations.                                                                                                                                                                                                                                                                                        |
| <b>Skills</b>               | Measuring dimensions of a frustum (top radius $R$ , bottom radius $r$ , slant height, vertical height $h$ ); substituting values into the frustum-of-cone formula; converting between $\text{cm}^3$ and litres; comparing capacities of different containers; communicating mathematical reasoning step by step.                                                                                                                                                                                                                                                         |
| <b>Attitudes</b>            | Appreciate the power of mathematical derivation in explaining everyday observations; show curiosity about why similar-looking containers hold different amounts; value precision in measurement and calculation; respect peers' reasoning during collaborative derivation.                                                                                                                                                                                                                                                                                               |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Purpose in Storyline</b> | In Lessons 1–3 learners established that a cone holds exactly one-third of a same-dimension cylinder, and calculated volumes of cones and pyramids. Lesson 4 now tackles the third container in the phenomenon — the metal frustum bucket. Learners physically measure the bucket, reconstruct the full cone from which it was cut, subtract volumes, and arrive at $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ . This is the mathematical climax that explains why the bucket holds more than the cone but less than the cylinder, fully resolving the anchoring phenomenon. |
| <b>Safety Notes</b>         | Ensure water is cleaned up immediately if spilled during hands-on measurement to prevent slipping. Handle metal bucket edges with care — inspect for sharp rims before passing to learners. Use non-breakable measuring containers only.                                                                                                                                                                                                                                                                                                                                 |

### B. LESSON OVERVIEW

Learners begin by predicting where the frustum bucket's capacity sits relative to the cone and cylinder using their prior knowledge. They then measure a real frustum-shaped bucket (top radius  $R$ , bottom radius  $r$ , vertical height  $h$ ) in groups, reconstruct the dimensions of the full cone from which it was cut using similar-triangle reasoning, and derive  $V(\text{frustum}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)$  step by step. Using this formula they calculate the bucket's capacity in litres, compare it to the cylinder and cone results from Lessons 1–3, and fully resolve the anchoring phenomenon. The lesson closes with learners revising their cumulative class model and Driving Question Board to capture the complete three-container relationship.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                   | Learner Experience                                                                                                                                             | Teacher Moves                                                                                                                                                | Sensemaking Strategy                                                                                                                                             | Formative Assessment Strategy                                                                                                                                         |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Volumes of cones intuition</a><br>Source: Khan | Each learner individually examines a diagram showing the three Kericho containers (cylinder, cone, frustum bucket) — all with the same height of 30 cm and the | Display the three-container diagram on the board (or hold up the physical containers). Say: 'Look at these three containers that Kamau and Atieno's tea farm | Anchored Prediction with Boundary Constraints — learners use the established cone ( $\frac{1}{3}$ cylinder) relationship as a lower bound and the cylinder as an | Observable indicator: Learner's written prediction includes (1) a numerical estimate between 6.16 L and 18.47 L and (2) a written justification referencing the cone- |

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| <p>Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Volume of the Cone (PLIX)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                            | <p>same top opening diameter of 28 cm — alongside a partially completed capacity table from Lessons 1–3. The table shows: Cylinder <math>\approx</math> 18.47 L, Cone <math>\approx</math> 6.16 L. The Frustum row is blank. Learners write a personal prediction: 'I predict the frustum bucket holds approximately ____ litres because ____.' They are also asked to sketch what the full cone might look like if the frustum were 'completed' downward, labelling where the tip of the cone would be.</p>                                                                                                                                                                                                                                                                                                                                                      | <p>uses in Kericho. We have calculated two of them. Today we settle the mystery of the bucket.' Hand out the prediction worksheet. Allow 3 minutes of silent individual thinking. After 3 minutes say: 'Write your prediction and your reason — even if you are not sure, commit to a number.' After writing, cold-call 4 learners to share predictions (select one who said 'closer to the cone', one 'closer to the cylinder', one 'exactly in the middle', and one who used a fraction argument). Record all four predictions visibly on the board under the heading 'OUR PREDICTIONS — we will check these at the end.' Do NOT confirm or reject any prediction. Say: 'Keep these in mind — your evidence today will tell you who was right.'</p>                                                                            | <p>upper bound to reason that the frustum must lie between the two values. This activates prior knowledge from Lessons 1–3 and creates cognitive need for the derivation.</p>                                                                                                                                                             | <p>cylinder relationship or the physical appearance of the frustum. Teacher scans 6–8 predictions during the 3-minute window. Flag any learner who predicts outside the 6–16 L range for targeted questioning in Phase 2.</p>                                                                                                                                                                            |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Volume of a cone</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume of cones</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Groups of 4 receive one metal bucket (frustum-shaped), a ruler, a tape measure, and a recording sheet. They measure: top inner diameter (D), bottom inner diameter (d), and vertical height h. They convert to radii: <math>R = D/2</math>, <math>r = d/2</math>. They record raw measurements and compare with two neighbouring groups. Then learners watch a 4-minute teacher-guided demonstration: the teacher places the frustum bucket next to a tall cone model (made from card) that has the same top radius and same slant angle. The teacher tilts the cone to show it could be 'completed' downward to a tip. Learners observe where the tip of the full cone would be relative to the bucket's bottom. They record: 'The full cone is taller than the bucket. The small cone cut away is shorter. Together, large cone = frustum + small cone.'</p> | <p>Distribute equipment and say: 'You have 6 minutes to measure your bucket carefully. Measure top diameter, bottom diameter, and vertical height. Record in your table. Each person in the group must independently record — do not share one answer.' Circulate and check at least 4 groups. After 6 minutes, cold-call 3 groups to report their R, r, h values. Write them on the board. If values differ by more than 0.5 cm ask: 'What might cause the difference? How do engineers handle measurement uncertainty?' (WAIT TIME: 12 seconds.) Then hold up the cone model. Say: 'This cone has the same opening — radius 14 cm — as the bucket. Watch what happens when I place it inside.' Slowly lower the cone into the frustum outline drawn on the board. Ask: 'What solid is left inside the cone but outside the</p> | <p>Concrete–Pictorial–Abstract (CPA) Sequence — learners first handle physical objects (concrete measurement), then observe the cone-within-frustum demonstration (pictorial/representational), and will move to algebra in Phase 3 (abstract). Naming the strategy aloud helps learners track where they are in the reasoning chain.</p> | <p>Observable indicator: Each group's recording sheet shows three distinct measurements (R, r, h) with units. Learners can verbally state 'Large cone = Frustum + Small cone' when cold-called. Teacher listens for incorrect identification of slant height as vertical height — correct immediately with: 'We need the perpendicular height, not the slant. How would you measure that carefully?'</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Metric system: units of volume</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Volume of Cones (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Guided derivation in pairs. Learners are given a printed diagram of the large cone (height H, radius R) with the small cone (height H–h, radius r) shaded inside it. They work through five scaffolded steps written on the board: Step 1 — Write V(large cone) in terms of H and R. Step 2 — Use similar triangles to show <math>H/R = (H-h)/r</math>, then solve for H to get <math>H = Rh/(R-r)</math>. Step 3 — Write V(small cone) in terms of (H–h) and r. Step 4 — Subtract: <math>V(\text{frustum}) = V(\text{large}) - V(\text{small}) = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 (H-h)</math>. Step 5 — Substitute <math>H = Rh/(R-r)</math>, expand, simplify algebraically to reach <math>V = \frac{1}{3}\pi h(R^2 + Rr + r^2)</math>. Pairs then substitute their measured bucket values to calculate volume in <math>\text{cm}^3</math> and convert to litres (<math>\div 1000</math>). They also calculate the percentage of the cylinder's capacity that the bucket holds.</p> | <p>Say: 'We observed that <math>V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})</math>. Now we will use algebra to turn that observation into a formula we can use for any frustum.' Write Step 1 on the board and pause. 'Write V(large cone). Everyone.' (WAIT TIME: 10 seconds.) Cold-call 1 learner for the answer. Affirm: 'Good — <math>\frac{1}{3}\pi R^2 H</math>. But we don't know H yet — that is what Step 2 solves.' Work through the similar-triangle argument carefully: draw the two right triangles on the board, label R, r, H, H–h. Ask: 'Why are these two triangles similar?' (WAIT TIME: 12 seconds.) Cold-call 2 learners — listen for 'same apex angle' or 'same slant direction.' After deriving <math>H = Rh/(R-r)</math>, say: 'This is the most important algebraic step. Copy it, box it.' Steps 4 and 5 — work one line at a time, pausing for pairs to verify each line. At Step 5 say: 'This formula — one-third pi h, bracket, R-squared plus R-r plus r-squared — is the formula for ANY frustum of a cone. Add it to your formula card.' Then say: 'Now substitute your group's measured values. Calculate the bucket's volume in <math>\text{cm}^3</math>, then convert to litres.' Allow 8 minutes for calculation. Cold-call 3 groups to share their answers. Circle back to the phenomenon: 'So where does the bucket sit — closer to the cone at 6.16 L or the cylinder at 18.47</p> | <p>Step-by-Step Algebraic Derivation with Sense-Checks — rather than presenting the formula as a rule to memorise, each algebraic step is derived and sense-checked against the physical model. This builds understanding of WHY the formula has three terms (<math>R^2</math>, <math>Rr</math>, <math>r^2</math>) — they arise naturally from expansion. The strategy explicitly connects symbolic manipulation to the physical phenomenon of 'cutting' a small cone from a large one.</p> | <p>Observable indicator: Learners' worked derivation in exercise books shows all 5 steps with correct algebraic manipulation, particularly the similar-triangle step <math>H = Rh/(R-r)</math> and the final simplified form. Teacher checks 5 books during the 8-minute calculation window. Success threshold: at least 4 of 5 books show a final volume between 10 L and 15 L for typical bucket dimensions (<math>R \approx 14</math> cm, <math>r \approx 9</math> cm, <math>h \approx 30</math> cm). Flag pairs whose answer is outside this range for immediate peer-check.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving similar triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Whole-class DQB update (10 minutes). The class Driving Question Board displays three columns: WHAT WE OBSERVED   WHAT WE NOW KNOW   WHAT WE STILL WONDER. Learners write new sticky-note contributions: column 2 entries include 'V(frustum) = <math>\frac{1}{3}\pi h(R^2 + Rr + r^2)</math>', 'Bucket holds <math>\approx [X]</math> L — more than cone, less than cylinder', 'The formula comes from subtracting cones — it has THREE terms because of <math>(R+r)(R^2 - Rr + r^2)</math> factoring... or from the expansion.' Column 3 entries (wonder questions) might include: 'Can we derive a formula for a frustum of a square pyramid the same way?', 'What if <math>R = r</math> — does the formula give a cylinder? Let's check!', 'How do engineers who design KPLC water tanks use this?'. The completed capacity table now shows all three containers: Cone 6.16 L → Frustum <math>\approx</math> 12.3 L → Cylinder 18.47 L.</p> | <p>Say: 'Take two sticky notes. On the first, write one thing you now KNOW that you didn't know at the start of this lesson — make it mathematical. On the second, write one question you still have or something you wonder about.' Allow 3 minutes of silent writing. Then invite learners to bring their notes to the board. Read 4 'know' notes and 4 'wonder' notes aloud. For one wonder note that says 'What if <math>R = r</math>?' say: 'Let's test it right now — substitute <math>R = r</math> into our formula.' (WAIT TIME: 15 seconds.) Cold-call 1 learner to work it on the board — the formula should collapse to <math>\frac{1}{3}\pi h(3r^2) = \pi r^2 h</math>, which is the cylinder formula. Say: 'Does that make sense? When the top and bottom radii are equal, a frustum IS a cylinder.' Point to the driving question on the board and say: 'Our driving question asks HOW we can design a container to hold exactly the capacity we need. What does today's formula give us the power to do?'</p> | <p>Know–Wonder Board Protocol — separating 'what we know' from 'what we still wonder' externalises metacognitive awareness and shows learners that mathematical understanding is cumulative and ongoing. The live check (<math>R=r</math> gives cylinder) models mathematical sense-making as verification through special cases, a key NGSS practice of 'Using Mathematics and Computational Thinking.'</p>                                                            | <p>Observable indicator: 'Know' sticky notes contain mathematically precise statements (formula written correctly, numerical result for the bucket stated in litres). 'Wonder' notes show genuine extension thinking beyond the lesson content. Teacher screens notes for misconceptions — e.g., a note saying 'frustum always holds half of the cylinder' is flagged and gently corrected publicly: 'That's an interesting conjecture — let's test it with our numbers. Does our bucket hold half of 18.47 L?'</p>                                                                                                               |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving similar triangles: same side plays different roles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Volume of the</a></p>                                                                                                                                                     | <p>Learners individually update their cumulative Capacity Model — a labelled diagram they have been building across Lessons 1–4. They add the frustum bucket with full dimensions, the derived formula, and the calculated capacity. They draw a three-container comparison bar (like a capacity bar chart) showing Cone : Frustum : Cylinder in correct proportion. They annotate: 'The frustum holds more than the cone because it has a wider base — it retains the bottom width of the large cone but</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Say: 'Open your Capacity Models from Lessons 1 to 3. We are adding our third and final container today.' Give learners 5 minutes to draw, label, and annotate. Circulate and check that the formula is written correctly and the bar chart proportions are roughly accurate (frustum bar should be about <math>\frac{2}{3}</math> of cylinder bar). After 5 minutes say: 'Now the exit task — this is individual. You have 7 minutes.' Read the prompt aloud once. Remind learners: 'Show every substitution step — do not</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Cumulative Model Revision — learners physically add to a model they have built over four lessons, making the growth of their understanding visible and tangible. The proportional capacity bar chart forces quantitative reasoning about relative sizes. The exit prompt applies the formula to a novel Kenyan context, assessing transfer. This reflects the NGSS practice of 'Developing and Using Models' as an ongoing scientific and mathematical activity.</p> | <p>Observable indicator (model): Frustum diagram is correctly labelled with R, r, h, formula, and capacity in litres. Bar chart shows frustum between cone and cylinder with approximately correct proportions. Observable indicator (exit prompt): Learner correctly substitutes <math>R=15</math>, <math>r=10</math>, <math>h=35</math> into <math>V = \frac{1}{3}\pi(35)(15^2 + 15 \times 10 + 10^2) = \frac{1}{3}\pi(35)(225 + 150 + 100) = \frac{1}{3}\pi(35)(475) \approx 17,495 \text{ cm}^3 \approx 17.50 \text{ L}</math>. Recommendation is justified by comparing to the class bucket's volume. Success threshold:</p> |

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| <a href="#">Cone (PLIX)</a><br>Source: CK-12<br><a href="#">Search ARES</a><br>for similar readings | discards the narrow tip where volume is least.' Finally, learners respond to the Exit Prompt in writing: 'A different tea farm in Kericho uses a frustum bucket with $R = 15$ cm, $r = 10$ cm, $h = 35$ cm. Calculate its capacity in litres to 2 decimal places. Would you recommend this bucket or the one we measured today for carrying more tea leaves? Justify your answer mathematically.' | just write the answer.' After 7 minutes, collect written responses. Before dismissing, point to the three containers one final time and say: 'At the start of this unit, Kamau and Atieno's farm workers were puzzled — same height, same opening, very different amounts. Can you now explain this to a Form 1 student in one sentence?' (WAIT TIME: 12 seconds.) Cold-call 2 learners for their one-sentence explanations. Close with: 'Next lesson we extend this to frustums of pyramids — the principle is identical, but the base shape changes.' |  | correct formula application and answer in range 17.40–17.60 L. Teacher reviews exit prompts before Lesson 5 to identify learners who need re-teaching of the similar-triangle H derivation. |
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#### D. TEACHER REFLECTION

After collecting exit prompts, review these three questions before Lesson 5:

1. FORMULA DERIVATION: What percentage of learners correctly wrote  $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$  without copying from a neighbour? If fewer than 70% did so independently, open Lesson 5 with a 5-minute paired re-derivation from scratch before moving to pyramid frustums.
2. SIMILAR-TRIANGLE STEP: Which learners struggled to accept  $H = Rh/(R-r)$ ? This algebraic manipulation requires comfort with literal equations. Prepare a visual proof using two nested cone diagrams for those learners.
3. PHENOMENON CONNECTION: Did learners' one-sentence closing explanations reference both shape (geometry) and quantity (formula/numbers)? A purely geometric explanation ('the cone has a pointy tip') without quantitative support suggests learners have intuition but not yet mathematical precision — plan a short 'convince me with numbers' challenge at the start of Lesson 5.

Also reflect: Was the 80-minute period sufficient for the full derivation AND the exit prompt? If the derivation ran long, consider separating Step 5 (algebraic expansion) into a structured homework task so that Lesson 4 class time focuses on measurement, the key subtraction insight, and the formula application.

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>What did I observe?</b>                   | Learners physically measured a frustum-shaped metal bucket (top radius $R$ , bottom radius $r$ , vertical height $h$ ) and observed a cone model placed alongside it, noticing that the frustum is a large cone with a small cone removed from the bottom.                                                                                                                                                                                                                                                                      |
| <b>What did I learn?</b>                     | Learners derived $V(\text{frustum of cone}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ by subtracting $V(\text{small cone})$ from $V(\text{large cone})$ , using similar-triangle reasoning to eliminate the unknown full height $H$ . They calculated the Kericho bucket's capacity in litres and placed it correctly between the cone (~6.16 L) and the cylinder (~18.47 L), resolving the anchoring phenomenon.                                                                                                                     |
| <b>How does this explain the phenomenon?</b> | The frustum bucket holds more water than the cone because removing the narrow tip of the cone discards very little volume — the tip contributes almost nothing — while retaining the wide bottom section. The three terms $R^2$ , $Rr$ , $r^2$ in the formula arise algebraically from expanding the subtraction of two cone volumes and reflect the interplay between the large radius, the small radius, and their geometric mean. This is why containers of the same height and top diameter can hold dramatically different |

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|  | amounts depending on their shape. |
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## LESSON 5 (80 minutes): Volume of Frustums of Pyramids — Designing the Perfect Grain-Storage Bin

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Purpose</b>              | To extend the frustum concept from cones to square and rectangular pyramidal frustums, and to apply the formula $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$ in computing the volume and capacity of a pyramidal grain-storage bin in litres and kilograms of maize.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Knowledge</b>            | The learner knows that: (a) a frustum of a pyramid is formed by cutting a pyramid with a plane parallel to its base; (b) the volume of a frustum of a pyramid is given by $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$ , where $A_1$ and $A_2$ are the areas of the two parallel faces and $h$ is the perpendicular height between them; (c) 1 litre = 1 000 cm <sup>3</sup> ; and (d) mass = volume × density, enabling conversion of capacity in litres to kilograms of maize (density ≈ 0.72 kg/litre).                                                                                                                                                                                                                                                                                                                            |
| <b>Skills</b>               | The learner is able to: (a) identify the two parallel faces and the perpendicular height of a pyramidal frustum; (b) compute the areas $A_1$ and $A_2$ for square and rectangular bases; (c) substitute correctly into $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$ and evaluate the expression step by step; (d) convert the volume in cm <sup>3</sup> to litres; (e) calculate the mass of maize the bin can hold using density; (f) update their cumulative DQB model to include pyramidal frustums alongside conical frustums, cylinders, and cones.                                                                                                                                                                                                                                                                              |
| <b>Attitudes</b>            | The learner appreciates that mathematical formulas are practical tools for solving real-world design challenges faced by Kenyan farmers, and values precision in calculation as a form of respect for the needs of the community.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Purpose in Storyline</b> | In Lessons 1–4 learners established that $V_{\text{cylinder}} = \pi r^2 h$ , $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$ , and $V_{\text{frustum(cone)}} = \frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1 r_2)$ . Today they discover that the same structural logic — the average of two base areas weighted by the geometric mean — extends to pyramidal frustums. By computing the capacity of a square-based pyramidal grain-storage bin used in Kericho and Rift Valley farms, and converting that capacity to kilograms of maize, learners move from the tea-leaf containers of the anchoring phenomenon to a broader class of frustum shapes, deepening their answer to the driving question: the shape of a frustum's cross-section (circular or rectangular) changes $A_1$ and $A_2$ but leaves the volume formula structurally identical. |
| <b>Safety Notes</b>         | No hazardous materials are used. If physical cardboard models are assembled, learners should take care with scissors and craft knives; these should only be handled under teacher supervision. Ensure adequate desk space so that 3-D models are not knocked over.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### B. LESSON OVERVIEW

Lesson 5 of 8 builds directly on Lesson 4's treatment of conical frustums. The anchoring phenomenon is revisited: the Kericho tea farmers' metal bucket (a conical frustum) is now compared to a square-based pyramidal bin used to store harvested maize in the Rift Valley. Learners first predict whether the pyramidal bin holds more or less than the conical frustum of similar dimensions, then observe a worked derivation linking the pyramid-frustum formula to the cone-frustum formula they already know. They explain and apply the formula  $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$  to a realistic grain-bin design problem, convert the result to litres and kilograms of maize, and finally update both their Driving Question Board and their cumulative cross-section model to include pyramidal frustums. The lesson reinforces the KICD-mandated competencies of critical thinking, creativity, and unity while embedding financial-literacy and food-security PCIs.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                           | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Volume density</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Volume of prisms and pyramids</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar readings | Learners are shown two side-by-side scale drawings on the board: (A) the Kericho tea-farmers' metal bucket from the anchoring phenomenon — a conical frustum with top radius $r_1 = 14$ cm, bottom radius $r_2 = 9$ cm, height $h = 30$ cm — and (B) a square-based pyramidal maize-storage bin with top square side $a_1 = 26$ cm, bottom square side $a_2 = 18$ cm, and the same height $h = 30$ cm. Learners individually write a prediction in their exercise books: 'Which container holds more, and by how much? Explain your reasoning in at least two sentences.' They then share with a shoulder partner for 2 minutes before a whole-class cold-call. | Display both drawings and say: 'Look carefully at these two containers. They have the same height — 30 centimetres. I want you to silently predict which holds more water and write down WHY you think so. You have 3 minutes — go.' [Allow 3 minutes of silent individual writing.] After writing, say: 'Turn to your neighbour and compare predictions. Do you agree or disagree? You have 2 minutes.' [Allow 2 minutes.] After partner talk, say: 'Let us hear some predictions. I will cold-call — Achieng, what did you and your partner predict?' [WAIT 10 seconds after naming before prompting.] Cold-call 4 learners (aim for at least 2 different predictions). Record all predictions verbatim on a section of the board labelled 'Our Predictions — Lesson 5'. Do NOT confirm or correct yet. End by saying: 'We are going to find out exactly which is larger — and by how much — by the end of today's lesson.' | Elicit-and-record predictions: by surfacing prior reasoning (area comparisons, mental estimation) before instruction, the teacher creates a cognitive anchor. Learners will compare their predictions against computed results at the end of the lesson, making the resolution of surprise a powerful sensemaking moment.                                                                                                                                      | Observable indicator: every learner has at least two written sentences in their book before partner talk. Teacher circulates and scans for (a) learners who compare base areas (strong prior knowledge), (b) learners who only compare heights (partial understanding), and (c) learners who are unsure how to start (need scaffolding during the Observe phase). Teacher notes names for targeted cold-calling later.                                                           |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Applying volume of solids</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Use related volumes</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan                                                                                                                                                                  | Learners observe a three-part structured derivation displayed on the board and in their ARES workbooks. Part A (5 min): Teacher displays a whole square pyramid and its frustum cut-out, labelling $A_1$ (top face area), $A_2$ (bottom face area), and $h$ (perpendicular height). Learners copy the diagram and label into their books. Part B (8 min): Teacher walks through the algebraic derivation of $V_{\text{frustum(pyramid)}} = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$ side by side with the cone-frustum formula $V = \frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1r_2)$ , showing that both                                                               | Part A: Sketch the pyramid frustum slowly. Say: 'This is a frustum of a square pyramid — the shape many grain-storage bins and maize silos in the Rift Valley use. What are the two parallel faces?' [WAIT 10 seconds.] Cold-call 2 learners. Label $A_1$ and $A_2$ . Say: ' $h$ is always the perpendicular distance between the two parallel faces — not the slant height.' Underline 'perpendicular' on the board. Part B: Write both formulas on the board simultaneously. Say: 'I want you to look at the cone-frustum formula we derived in Lesson 4 — here — and the                                                                                                                                                                                                                                                                                                                                                   | Side-by-side formula comparison with colour-coding: placing the cone-frustum and pyramid-frustum formulas next to each other and having learners physically colour matching terms makes the structural analogy visible and tactile, reducing cognitive load and building the generalisation that the same mathematical skeleton underlies both. This directly advances the driving question: both the bucket and the pyramidal bin obey the same volume logic. | Observable indicators: (a) learners' annotations show correct colour-coding of matching terms in both formulas; (b) in Part C, learners who correctly compute $\sqrt{A_1A_2} = 240$ cm <sup>2</sup> are on track — teacher checks this intermediate step explicitly during circulation; (c) thumbs-up count after Part C gives a rapid whole-class gauge. If fewer than 70% of learners show thumbs-up, the teacher re-models the $\sqrt{A_1A_2}$ step before moving to Phase 3. |

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| <p>Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>share the identical structure: <math>\frac{1}{3} \times h \times (\text{top area} + \text{bottom area} + \text{geometric mean area})</math>. Learners annotate: they colour-code matching terms in both formulas with two different coloured pens. Part C (7 min): Learners independently work through a guided numerical example in their ARES workbook — a pyramidal bin with <math>a_1 = 20</math> cm, <math>a_2 = 12</math> cm, <math>h = 25</math> cm — filling in a structured calculation scaffold with blanks for <math>A_1</math>, <math>A_2</math>, <math>\sqrt{A_1 A_2}</math>, the sum, and the final volume.</p>                                                | <p>pyramid-frustum formula — here. What do you notice about their structure?' [WAIT 12 seconds.] Cold-call 3 learners. Guide learners to see the structural identity. Say: 'The formula does not care whether the base is a circle or a square — it always says: one-third times height times the sum of the top area, the bottom area, and the geometric mean of the two areas. Write that sentence in your own words under the formulas.' Part C: Say: 'Now try the guided example in your workbook on your own — you have 5 minutes. Show every step.' Circulate, checking that learners correctly compute <math>A_1 = 20^2 = 400</math> cm<sup>2</sup>, <math>A_2 = 12^2 = 144</math> cm<sup>2</sup>, <math>\sqrt{A_1 A_2} = \sqrt{(400 \times 144)} = \sqrt{57\,600} = 240</math> cm<sup>2</sup>, sum = 784 cm<sup>2</sup>, <math>V = \frac{1}{3} \times 25 \times 784 = 6\,533.3</math> cm<sup>3</sup>. After 5 minutes: 'Who got 6 533 point 3 cubic centimetres? Show me a thumbs-up.' Address the most common error observed.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Volume density</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b></p> <p><a href="#">Volume of Pyramids: Fluctuating Height (PLIX)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>Learners work in groups of four to solve the main application problem, which is explicitly connected to the anchoring phenomenon: 'A maize farmer in Kericho uses a pyramidal grain-storage bin with a square top of side 30 cm, a square bottom of side 20 cm, and a perpendicular height of 40 cm. (a) Calculate the volume of the bin in cm<sup>3</sup>. (b) Convert the volume to litres. (c) The farmer plans to fill the bin with dry maize. Given that dry maize has a density of approximately 0.72 kg per litre, calculate the mass of maize the full bin can hold. (d) The farmer wants to double the capacity. If the height stays the same, what side length</p> | <p>Distribute the A3 problem sheet. Say: 'This is the same Kericho farming community from our anchoring phenomenon — now storing maize, not tea leaves. Work in your groups. Assign roles: one calculator, one recorder, one checker, one presenter. You have 15 minutes.' [Circulate actively. After 5 minutes, check that all groups have correctly computed <math>A_1 = 30^2 = 900</math> cm<sup>2</sup> and <math>A_2 = 20^2 = 400</math> cm<sup>2</sup>. If a group is stuck on part (d), ask: 'In Sub-Strand 2.1 we found that if the volume scale factor is <math>k^3</math>, and we want volume to double, what must <math>k^3</math> equal?' WAIT 12 seconds before offering further guidance.] After 15 minutes, say: 'Group 3, please present your</p>                                                                                                                                                                                                                                                                          | <p>Collaborative problem-solving with cross-group critique: having two groups present — one correct, one with an error — turns the comparison into a sensemaking activity where learners must apply their emerging understanding to evaluate claims. The return to the opening predictions closes a predict-observe-explain loop within the lesson, directly connecting to the anchoring phenomenon and the driving question. Connecting part (d) to Sub-Strand 2.1 (Similarity and Enlargement) makes cross-strand mathematical coherence visible.</p> | <p>Expected answers for formative checking: (a) <math>A_1 = 900</math> cm<sup>2</sup>, <math>A_2 = 400</math> cm<sup>2</sup>, <math>\sqrt{A_1 A_2} = 600</math> cm<sup>2</sup>, <math>V = \frac{1}{3} \times 40 \times 1900 = 25\,333.3</math> cm<sup>3</sup>; (b) 25.33 litres; (c) <math>25.33 \times 0.72 \approx 18.24</math> kg of maize; (d) <math>k = \sqrt[3]{2} \approx 1.26</math>, new top side <math>\approx 37.8</math> cm, new bottom side <math>\approx 25.2</math> cm. Teacher uses the A3 sheets as a visible formative record — collect all sheets at the end of class for written feedback before Lesson 6.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>should both the top and the bottom square be scaled to? (Use the volume scale factor from Sub-Strand 2.1 — Similarity and Enlargement.)' Each group records their working on an A3 sheet. After 15 minutes of group work, two groups present their solutions to the class (teacher selects one group with a correct solution and one group that made a common error, without pre-announcing which is which). The class critiques both presentations.</p>       | <p>working for parts (a), (b) and (c).' After Group 3 presents, say: 'Group 6, please present your working for part (d).' After both presentations, pose to the class: 'Do you agree with Group 3's answer for part (c)? Raise your hand if you got the same mass of maize.' Cold-call one learner who disagrees to explain the discrepancy. For part (d), explicitly ask: 'If volume doubles, <math>k^3 = 2</math>, so <math>k = \sqrt[3]{2} \approx 1.26</math> — what does that tell us about the new side lengths?' Cold-call 2 learners. Confirm: new top side <math>\approx 30 \times 1.26 \approx 37.8</math> cm, new bottom side <math>\approx 20 \times 1.26 \approx 25.2</math> cm. Close by saying: 'Now return to our predictions from the start of the lesson. Was the pyramidal bin larger or smaller than the conical frustum? Let us check.' [Work through <math>V_{\text{cone-frustum}} = \frac{1}{3}\pi(30)(15^2 + 10^2 + 150) = \frac{1}{3}\pi(30)(400) = 12\,566 \text{ cm}^3</math> vs <math>V_{\text{pyramid-frustum}} = \frac{1}{3}(40)(900 + 400 + 600) = \frac{1}{3}(40)(1900) = 25\,333 \text{ cm}^3</math>.] Say: 'The pyramidal bin holds MORE — about twice as much. How many of you predicted that? Why does it hold more even though both have the same height?'</p> |                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b> <a href="#">Applying volume of solids</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> | <p>Each learner independently writes on a sticky note (or in a designated DQB section of their workbook if physical sticky notes are unavailable): one thing they can NOW explain about the driving question that they could not explain before today, and one question that today's lesson has raised for them. Learners post or write their notes. The teacher reads out 4–5 contributions and clusters them on the DQB under two growing columns: 'What we</p> | <p>Say: 'Take a sticky note — or open your workbook to the DQB page. Write: one thing you can NOW explain about why the tea farmers' containers hold different amounts, and one new question today has made you think about. You have 3 minutes — on your own, silent.' [WAIT 3 minutes.] 'Post your notes. I am going to read some aloud.' Read 4–5 notes, grouping them visibly on the board. Highlight any note that connects to the driving question — e.g., 'The</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>DQB update with explicit tracking table: making cumulative understanding visible on a shared board (and in individual workbooks) gives learners a metacognitive map of the unit's evidence-gathering sequence. Clustering learner-generated questions around the driving question sustains inquiry momentum and shows learners that their confusion is productive and shared.</p> | <p>Observable indicator: each learner's sticky note or workbook entry contains (a) a specific mathematical claim (not a vague statement like 'I learned about frustums') and (b) a genuine question. Teacher scans for entries that reveal lingering misconceptions — e.g., 'Is the formula different if the base is a triangle?' — and flags these for targeted attention in Lesson 6.</p> |

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| <p><a href="#">Surface Area and Volume of Pyramids Practice</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>can now explain' and 'What we still need to find out'. Learners update the DQB tracking table in their workbooks, ticking off: cylinder ✓, cone ✓, conical frustum ✓, pyramidal frustum ✓ (today), and noting that 'designing a container of exact capacity' remains the culminating challenge.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>formula is the same structure for any frustum — it does not matter if the base is a circle or a rectangle.' Say: 'That is a big idea. The Kericho tea farmers' bucket and the Rift Valley maize bin both obey the same mathematical rule — one-third times height times that special sum of areas. Tick pyramidal frustum in your tracking table. What is left for us to answer before we can fully design our own container?' Cold-call 2 learners. Guide towards: 'We need to know how to choose dimensions to hit an exact target capacity — that is our culminating design challenge in Lesson 8.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Surface Area and Volume of Pyramids (PLIX)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to their cumulative cross-section model — a side-by-side annotated diagram that has been built across Lessons 1–5. Today they add a new panel: a correctly labelled cross-section of a square-based pyramidal frustum, showing <math>A_1</math> (top face), <math>A_2</math> (bottom face), <math>h</math> (perpendicular height), and the formula <math>V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})</math>. Below the diagram they write: 'Structural link: this formula has the same form as the conical frustum formula because both express the average of two base areas weighted by their geometric mean, scaled by <math>\frac{1}{3}h</math>.' Learners also annotate an arrow connecting the pyramidal frustum panel back to the Similarity &amp; Enlargement panel, showing how the volume scale factor (<math>k^3 = 2 \rightarrow k = \sqrt[3]{2}</math>) was used in today's design question. Finally, they write one sentence connecting today's work back to the anchoring phenomenon: 'The Rift Valley pyramidal grain bin and the Kericho tea-leaf bucket are both frustums — their volumes</p> | <p>Say: 'Open your cumulative model — the diagram you have been building since Lesson 1. Today we add the pyramidal frustum panel. Draw it, label it, and write the formula. You have 6 minutes.' [Circulate and check that learners label <math>A_1</math> and <math>A_2</math> on the CORRECT faces — the two parallel faces — not the slant faces. Common error: labelling slant height as <math>h</math>. If observed, say: 'Remember — <math>h</math> is always perpendicular, straight down, between the two flat faces. Is your <math>h</math> pointing straight down?' WAIT 10 seconds.] After 6 minutes: 'Now write the structural link sentence and the arrow to Similarity and Enlargement. 3 minutes.' After 3 minutes: 'Finally, write your one sentence connecting today to the Kericho phenomenon. 2 minutes.' Close: 'Hold up your cumulative model. You now have five panels — cylinder, cone, conical frustum, pyramidal frustum, and the structural link. Three more lessons and you will use all of this to</p> | <p>Cumulative annotated model building: requiring learners to physically add each lesson's content to a single growing diagram externalises the unit's conceptual progression. The cross-strand annotation arrow (to Similarity &amp; Enlargement) makes mathematical coherence visible. The phenomenon-connection sentence ensures that every lesson phase closes the loop back to the anchoring context, sustaining the inquiry narrative across all eight lessons.</p> | <p>Observable indicators: (a) the pyramidal frustum panel shows <math>A_1</math> and <math>A_2</math> on the parallel faces (not slant faces) and <math>h</math> as perpendicular height; (b) the structural link sentence correctly uses the phrase 'geometric mean' or equivalent; (c) the phenomenon-connection sentence specifically names 'the Kericho tea-leaf bucket' or 'the Rift Valley grain bin' — generic sentences are flagged for revision. Teacher photographs 6–8 models (a purposive sample including strong and weak examples) for whole-class sharing at the start of Lesson 6.</p> |



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|  | differ because their base shapes differ, but the same formula structure governs both.' | design your own container. Well done.' |  |  |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. **PREDICTION QUALITY:** Did learners' opening predictions reveal sound reasoning about base areas, or were most relying only on height? If learners compared base areas spontaneously, this suggests strong transfer from Sub-Strand 2.1 (Similarity and Enlargement) — note this for the Lesson 6 introduction.
2. **FORMULA STRUCTURE:** When learners colour-coded the two formulas side by side, did they articulate the 'geometric mean' idea independently, or did it require teacher prompting? If it required prompting, plan an additional 5-minute consolidation at the start of Lesson 6.
3. **PART (d) — VOLUME SCALE FACTOR:** Was the cross-strand connection to Similarity and Enlargement ( $k^3 = 2 \rightarrow k = \sqrt[3]{2}$ ) accessible to most groups, or did it cause significant difficulty? If more than one-third of groups could not start part (d), consider a brief whole-class worked example at the beginning of Lesson 6 before moving to pyramids with non-square bases.
4. **DQB ENTRIES:** Review the sticky notes or workbook entries. Are learners' 'new questions' converging on the culminating design challenge (choosing dimensions for a target capacity), or are there unexpected gaps (e.g., confusion about what 'perpendicular height' means in a frustum)? Adjust Lesson 6 accordingly.
5. **MODEL BUILDING:** Were the  $A_1/A_2$  labelling errors (labelling slant faces instead of parallel faces) widespread? If yes, begin Lesson 6 with a 3-minute 'spot the error' activity using a photograph of an incorrectly labelled model.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>What did I observe?</b>                   | Learners observed a side-by-side derivation comparing the volume formula for a pyramidal frustum, $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{(A_1A_2)})$ , with the conical frustum formula from Lesson 4, revealing that both share an identical mathematical structure — one-third times height times the sum of the top area, the bottom area, and their geometric mean. They also observed scale drawings of the Kericho tea-leaf bucket (conical frustum) and a Rift Valley grain-storage bin (pyramidal frustum) with the same height but different base shapes.                                                                                                                                                                                                                           |
| <b>What did I learn?</b>                     | Learners determined that the volume of a frustum of a square or rectangular pyramid is given by $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{(A_1A_2)})$ , where $A_1$ and $A_2$ are the areas of the two parallel faces and $h$ is the perpendicular height. They computed that a pyramidal grain-storage bin (top side 30 cm, bottom side 20 cm, height 40 cm) has a volume of approximately $25\,333\text{ cm}^3 \approx 25.33$ litres and can hold approximately 18.24 kg of dry maize (density 0.72 kg/litre). They also applied the volume scale factor $k^3 = 2$ (from Sub-Strand 2.1) to determine that doubling the capacity while keeping the height fixed requires scaling each side length by $\sqrt[3]{2} \approx 1.26$ .                                                             |
| <b>How does this explain the phenomenon?</b> | Learners explained that the Kericho tea farmers' metal bucket (conical frustum) and the Rift Valley pyramidal grain bin hold very different amounts of water even at the same height because their base areas — and therefore the values of $A_1$ , $A_2$ , and $\sqrt{(A_1A_2)}$ — are fundamentally different: circular bases are governed by $\pi r^2$ , while square bases are governed by $s^2$ . The formula structure is the same, but the base-area calculation changes depending on the shape of the cross-section, which is why containers that look similar in height can hold dramatically different volumes. This advances the driving question: to design a container of exact capacity, a farmer must choose the base shape and dimensions carefully, not just the height. |



## LESSON 6 (80 minutes): Problem Solving with Real Containers: Apply & DQB Update

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Purpose</b>              | Learners apply volume and capacity formulas for cones, cylinders, and frustums to solve authentic Kenyan-context word problems, justify formula choices, and consolidate understanding of why the Kericho tea farmers' containers hold such different volumes despite similar appearances.                                                                                                                                                                                                                                                                                               |
| <b>Knowledge</b>            | Learners demonstrate knowledge of the volume formulas for cones ( $V = \frac{1}{3}\pi r^2 h$ ), cylinders ( $V = \pi r^2 h$ ), frustums of cones ( $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ ), and frustums of pyramids; the relationship 1 litre = 1000 cm <sup>3</sup> ; and how container shape — not just apparent size — determines capacity.                                                                                                                                                                                                                                         |
| <b>Skills</b>               | Learners apply volume and capacity formulas to multi-step word problems; select appropriate formulas given a described container shape; convert between cm <sup>3</sup> and litres; verify reasonableness of answers against the anchoring phenomenon; and communicate mathematical reasoning to peers.                                                                                                                                                                                                                                                                                  |
| <b>Attitudes</b>            | Learners appreciate that mathematical precision in calculating volume and capacity has direct practical value in everyday Kenyan life — from sizing water storage tanks in Turkana to designing cooking vessels for communal githeri — and show respect for peers' solution strategies by listening actively during group presentations.                                                                                                                                                                                                                                                 |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Purpose in Storyline</b> | In Lessons 1–5, learners established the cylinder–cone volume relationship (3:1), derived the frustum formula by subtraction of cones, and measured the actual Kericho tea containers. Lesson 6 is the application and consolidation lesson: learners transfer all accumulated knowledge to three new real-world Kenyan problems (a Turkana water tank, a githeri pot, and a drum-filling count), present and defend their solutions, and update the Driving Question Board with mature, formula-backed explanations — bringing the unit close to a full answer to the driving question. |
| <b>Safety Notes</b>         | No hazardous materials are used. If physical containers and water are used for verification, place absorbent cloths on desks and ensure learners handle cylindrical drums carefully to avoid tipping. Learners with mobility considerations should be seated where group materials are easily accessible.                                                                                                                                                                                                                                                                                |

### B. LESSON OVERVIEW

Learners work in groups on a structured set of three Kenyan-context word problems that require them to select and apply volume/capacity formulas for cones, cylinders, and frustums. Problem 1 asks groups to size a conical water-storage tank for a homestead in Turkana; Problem 2 requires calculating the capacity of a frustum-shaped cooking pot needed to prepare githeri for 100 people at a community event in Kisumu; Problem 3 determines how many frustum-shaped cups (like the metal tea-collection buckets from Kericho) are needed to fill a cylindrical storage drum. Groups present solutions on the board, justify their formula choices by connecting back to the anchoring phenomenon, and the class updates the Driving Question Board with consolidated understanding. The lesson ends with a teacher-led synthesis that directly answers the driving question using evidence built across all six lessons.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Challenging Problems Involving Capacity in Liters - Word Problems</a><br>Source: CK-12<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Volume of cylinders, spheres, and cones word problems</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Each learner individually reads the three problem stems (displayed on the board / ARES screen) without any numbers filled in, and writes a one-sentence prediction for EACH problem: (a) Which formula will be needed? (b) Will the answer be larger or smaller than the Kericho cylindrical drum's capacity (established in Lesson 4 as approximately 18.5 litres)? Learners commit their predictions in their exercise books before groups are formed. This low-stakes individual commitment activates prior schema and creates personal investment in the group work that follows.                                                                          | Display the three problem stems with blanks where numbers will appear. Say: 'Before I give you the full numbers, I want each of you — on your own, in silence — to predict which formula you will need for each container shape and whether the answer will be bigger or smaller than our Kericho drum. You have 3 minutes. Write your prediction now.' Set a visible timer. After 3 minutes, cold-call 3 learners (one per problem) using name sticks: 'Achieng, which formula do you predict for Problem 1, and why?' Apply 12-second WAIT TIME before accepting the answer. After each response, ask a second learner: 'Kamau, do you agree with Achieng's prediction? What is your reasoning?' Do NOT confirm or correct at this stage — say: 'Hold that prediction. Your group work will tell you whether you were right.' | Prediction with Formula Identification — By requiring learners to name the formula before seeing numbers, this strategy forces retrieval of the cone, cylinder, and frustum formulas from memory and connects them to shape descriptions. It surfaces any residual misconceptions (e.g., using the cone formula for a frustum) that the group work will then resolve.                                                                                                                  | Circulate and scan exercise books during the 3-minute prediction window. Observable indicators of readiness: learner writes a recognisable formula (even partially, e.g., ' $\frac{1}{3}\pi r^2 h$ ') next to each problem stem. Flag any learner who writes the same formula for all three shapes — seat them in a group with a strong peer who can model formula selection during Problem Set work.                                                                        |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Cylinder volume &amp; surface area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Apply volume of solids</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a>                        | Groups of 4 receive the full Problem Set card. PROBLEM 1 (Cone — Turkana): A family in Turkana wants to build an underground conical water-storage tank. The tank must hold at least 500 litres of water. The available excavation depth (height) is 120 cm. What minimum base radius must the tank have? Give your answer to 1 decimal place. PROBLEM 2 (Frustum of cone — Githeri pot): A community kitchen in Kisumu is preparing githeri for 100 people. Each person needs 1.5 litres of githeri. The cooking pot is a frustum of a cone with a top (open) radius of 35 cm, a base radius of 20 cm, and a height of 50 cm. (a) Calculate the volume of the | Distribute Problem Set cards. Say: 'You have 25 minutes. Every member must be able to explain any step — I will cold-call individuals, not just the group reporter. Show every substitution step; do not skip lines.' Set a visible 25-minute timer. Circulate in a deliberate loop — visit each group at least twice. At the 8-minute mark, if any group is stuck on Problem 1, prompt: 'You need at least 500 litres. Convert that to $\text{cm}^3$ first. What is 1 litre in $\text{cm}^3$ ?' (10-second WAIT TIME). At the 15-minute mark, prompt Problem 2 groups who are only computing volume without answering part (b): 'You have the pot volume. Now how many pots? What operation                                                                                                                                    | Structured Problem Decomposition — The five-step scaffold (identify shape → write formula → substitute → calculate → connect to phenomenon) is an explicit sensemaking routine. It prevents learners from pattern-matching numbers into a memorised formula without understanding and ensures every calculation is tethered to the physical reality of the Kericho containers. The 'connect to phenomenon' step in particular bridges procedural fluency and conceptual understanding. | While circulating, look for three specific indicators: (1) Does the group correctly write $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ for Problems 2 and 3 — not $V = \pi r^2 h$ ? (2) Does the group correctly convert 500 litres → 500 000 $\text{cm}^3$ in Problem 1? (3) Does the phenomenon-connection sentence go beyond 'the drum holds more' to reference the mathematical ratio? Groups not meeting indicator (1) need targeted intervention before presentations begin. |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>pot in litres. (b) How many full pots must be cooked to feed all 100 people? PROBLEM 3 (Frustum cup vs Cylinder drum — Kericho link): The metal tea-collection cup used in Kericho is a frustum of a cone: top radius 9 cm, base radius 6 cm, height 14 cm. The cylindrical storage drum at the collection centre has radius 30 cm and height 50 cm. How many full cups of tea leaves must be emptied into the drum to fill it completely? Groups work through each problem, showing all steps: (i) identify the shape and write the formula, (ii) substitute values, (iii) calculate, (iv) state the answer with units, (v) write one sentence connecting the answer back to the anchoring phenomenon.</p> | <p>do you need?' For Problem 3, if groups forget to link back to the phenomenon, prompt: 'Your answer tells you how many cups fill the drum. Does this match what you predicted about the cylinder holding more than the cone? Explain.' In the final 2 minutes, instruct each group to choose a reporter AND a 'challenger' (a different person who must be ready to answer one question about the working).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume word problem: water tank</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume and surface area of cylinders</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Three groups (one per problem) present their full solution at the board — the reporter writes the working while the challenger stands ready. The rest of the class acts as a 'peer review panel': they must write one agreement and one probing question in their exercise books before the Q&amp;A opens. After each presentation, the teacher facilitates 3 minutes of structured peer questioning. Following all three presentations, the teacher leads a 5-minute whole-class synthesis comparing the three container capacities side by side and re-anchoring every result to the Kericho phenomenon.</p>                                                                                              | <p>Invite Group A to present Problem 1. After they write the solution, cold-call their challenger: 'Otieno, explain why you divided 500 000 by (<math>\frac{1}{3}\pi \times 120</math>) and not by (<math>\pi \times 120^2</math>).' WAIT 12 seconds. Open the floor: 'Panel — who has a probing question for Group A?' Accept 2 questions maximum. Repeat for Groups B and C. For the synthesis, draw a three-column comparison table on the board: Container   Formula used   Capacity (litres). Populate it live with class answers. Ask: 'Looking at this table — does anyone see a pattern between which formula gives the largest capacity for similar dimensions?' Cold-call 2 learners. Then say: 'Now look back at your prediction from the start of the lesson. Turn to your neighbour and tell them in one sentence whether your prediction was correct and what you understand now that you did not understand</p> | <p>Claim–Evidence–Reasoning (CER) Oral Presentation — Each presenting group must articulate a claim (the answer), cite evidence (the substituted formula and calculated value), and provide reasoning (the connection to the anchoring phenomenon). The peer-review panel structure makes the audience cognitively active rather than passive, and the three-column synthesis table makes the comparative pattern visible — directly addressing the driving question about why similar-looking containers hold different amounts.</p> | <p>During presentations, listen for: (1) Does the reporter use correct mathematical vocabulary — 'slant height', 'radius', 'frustum', 'capacity in litres'? (2) Does the challenger's explanation reveal understanding of formula derivation (e.g., 'the frustum formula comes from subtracting a small cone from a large cone')? (3) Do peer-panel questions probe conceptual understanding (e.g., 'Why did you use R and r, not just one radius?') rather than only arithmetic checks? Record names of learners who demonstrate depth of reasoning for extension tasks.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Solid geometry word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>The class revisits the Driving Question Board that has been built across Lessons 1–5. Each learner receives two sticky notes: one GREEN ('I can now explain...') and one ORANGE ('I still wonder...'). Learners write their notes individually (2 minutes), then one representative from each group places their group's GREEN note under the relevant column on the DQB (Cone, Cylinder, Frustum, or General Pattern). The teacher reads aloud three selected GREEN notes and two ORANGE notes, and the class votes (thumbs up/sideways/down) on whether each GREEN note fully answers part of the driving question. ORANGE notes are stored for Lesson 7 (model making and measurement).</p> | <p>Distribute sticky notes. Say: 'Two minutes — write on GREEN: one thing you can now explain about the Kericho farmers' containers that you could not explain at the start of this unit. Write on ORANGE: one thing you still genuinely wonder about.' After 2 minutes, invite groups to place GREEN notes. Read aloud: 'This group writes — [read note]. Class, thumbs: does this fully answer part of our driving question, or is there still a gap?' For each ORANGE note, say: 'We will carry this question into Lesson 7. I want you to hold onto this curiosity.' Before closing the DQB segment, point to the driving question poster and say: 'Based on today's problems — the Turkana tank, the githeri pot, the Kericho cup and drum — can someone give me the most complete answer we have so far to our driving question? You have 15 seconds to think, then I will cold-call.' WAIT 15 seconds. Cold-call 2 learners.</p> | <p>DQB Progressive Revision — The DQB is not a display but a living epistemic record. By physically moving notes and voting on the completeness of explanations, learners engage in metacognitive monitoring: they distinguish between procedural answers ('I can calculate the frustum volume') and conceptual answers ('I can explain why the frustum holds more than the cone because it retains the wide middle section that the cone discards'). This distinguishes surface-level from deep understanding.</p> | <p>Collect GREEN and ORANGE notes at the end of the lesson. Review overnight for two indicators: (1) GREEN notes that reference only a formula (not an explanation) signal procedural-only understanding — these learners need conceptual prompting in Lesson 7. (2) ORANGE notes that raise design questions ('How do we choose the radii to get exactly 500 litres?') indicate readiness for extension work on the frustum design task in Lesson 8.</p>                                                                                      |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Challenging Problems Involving Capacity in Liters - Word Problems</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Challenging Problems Involving</a></p>                                                                                                                                                                                                     | <p>Learners return to their running 'Container Capacity Model' — a diagram started in Lesson 1 showing the three Kericho containers side by side. They now add a fourth annotated layer to their model: (a) Label each container with its formula. (b) Add a 'capacity bar' drawn to scale next to each container (1 cm = 10 litres), based on the actual values calculated or measured across the unit. (c) Write a two-sentence 'Model Update Statement' below the diagram explaining why the cylindrical drum holds exactly</p>                                                                                                                                                                | <p>Say: 'Open your Container Capacity Model from Lesson 1. You have 7 minutes to add the three annotations I have written on the board. This is your cumulative record of what you have learned — make it clear enough that a fellow learner who missed this unit could understand it.' Write the three annotation tasks on the board. Circulate and prompt: 'Your capacity bar for the drum should be three times as long as the bar for the cone — does yours show that? If not, recalculate and adjust.' At 6 minutes, say: 'Share with your</p>                                                                                                                                                                                                                                                                                                                                                                                     | <p>Scaled Visual Modelling — Drawing the capacity bar to scale transforms an abstract numerical answer into a spatial, proportional representation. This directly mimics the 'same apparent size, different volume' puzzle of the anchoring phenomenon and makes the 3:1 cylinder-to-cone ratio perceptually obvious. The Model Update Statement requires learners to translate the visual back into language, reinforcing dual coding of the mathematical concept.</p>                                             | <p>Collect models at the end of class (or photograph them). Check for three specific indicators: (1) All three formulas are correctly labelled. (2) The capacity bars are approximately in the ratio cone : frustum : cylinder = 1 : ~1.5–2.5 : 3 (exact ratio depends on frustum dimensions used). (3) The Model Update Statement uses mathematical vocabulary ('one-third', 'frustum lies between', 'same base and height') and references the Kericho phenomenon by name. Learners whose bars are all equal length have not transferred</p> |

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| <a href="#">Capacity in Liters - Word Problems Practice</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | three times the conical cover, and where the frustum-shaped bucket sits between them mathematically. Learners share their updated model with a partner and identify one difference between their two models. | partner. Find ONE thing your partner's model shows that yours does not.' After 1 minute of partner sharing, cold-call 2 pairs: 'Wanjiku, what did you learn from your partner's model?' WAIT 10 seconds. Close by pointing to the driving question: 'Your model now visually answers why those three Kericho containers look the same size but hold such different amounts. You built that understanding across six lessons — that is the power of mathematical reasoning.' | the 3:1 insight and need a brief one-on-one check at the start of Lesson 7. |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Problem Set work, did all groups correctly identify the frustum formula for Problems 2 and 3 without prompting — or did most groups default to the cylinder formula? If more than two groups used the wrong formula, the formula-identification scaffold needs to be made more explicit in Lesson 7 (e.g., a 'shape identification checklist' card). (2) During presentations, did challengers demonstrate conceptual understanding of where the frustum formula comes from, or did they only reproduce the calculation steps? If challengers could not explain the derivation, plan a 5-minute 'formula origin' retrieval activity at the start of Lesson 7. (3) Were the ORANGE sticky notes largely procedural ('I still wonder how to calculate') or conceptual/design-oriented ('I still wonder how to choose dimensions to hit a target capacity')? The ratio of procedural to design-oriented ORANGE notes will tell you whether learners are ready for the open-ended design challenge in Lesson 8. (4) Did the 80-minute period allow genuine group discussion and full presentations, or were you rushed during the DQB and model-building phases? If time was tight, consider assigning the Model Update Statement as a written homework task so that the DQB update receives full class attention.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>What did I observe?</b>                   | Three different containers used by Kericho tea farmers — a cone, a cylinder, and a frustum-shaped metal bucket — appear nearly the same size when placed side by side, yet when filled with water the cylinder holds three times as much as the cone, and the frustum holds an amount between the two.                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>What did I learn?</b>                     | The volume of a cone is exactly one-third the volume of a cylinder with the same base radius and height ( $V = \frac{1}{3}\pi r^2 h$ vs $V = \pi r^2 h$ ). A frustum of a cone has volume $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ , which is always greater than the cone alone but less than the full cylinder. These formulas allow us to calculate and compare the exact capacity of any container — including sizing a Turkana water tank, a Kisumu githeri pot, or counting how many Kericho tea cups fill a storage drum.                                                                                                                                                                     |
| <b>How does this explain the phenomenon?</b> | The three Kericho containers look similar because they share roughly the same height and opening diameter, but their shapes distribute that space very differently. The cylinder fills every cross-section from base to top with a full circle of area $\pi r^2$ . The cone tapers to a point, so most cross-sections are much smaller — it uses only one-third of the cylinder's space. The frustum keeps a wide base and a wide top, so it retains far more of the middle volume than the cone does, placing its capacity between the cone and the cylinder. By applying the correct formula to each shape, we can predict, compare, and design containers to hold exactly the capacity we need. |





## LESSON 7 (80 minutes (double lesson)): Project: Model Making & Measurement — Frustum, Pyramid Frustum, and Cone Containers

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Purpose</b>              | Learners design and physically construct model containers shaped as a frustum of a cone, a frustum of a pyramid, and a cone using locally available materials (cardboard from cereal boxes, old packaging). They measure actual dimensions of their models, calculate theoretical volumes and capacities, then verify experimentally by filling with water or sand — connecting every measurement back to the Kericho tea farmers' three containers that anchor the whole unit.                                                                                                                                                                                                |
| <b>Knowledge</b>            | Learners demonstrate knowledge of the volume formulas for a cone ( $V = \frac{1}{3}\pi r^2 h$ ), a frustum of a cone ( $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ ), and a frustum of a pyramid ( $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})$ ), and understand that 1 litre = 1000 cm <sup>3</sup> for converting between volume and capacity.                                                                                                                                                                                                                                                                                                                                 |
| <b>Skills</b>               | Learners measure physical dimensions (radii/side lengths of top and bottom faces, slant height, perpendicular height) using a ruler or tape measure; calculate theoretical volumes and capacities using correct formulae; fill models with water or sand to obtain experimental capacity; compute percentage discrepancy between theoretical and experimental values; and document the process using photographs or short video clips.                                                                                                                                                                                                                                         |
| <b>Attitudes</b>            | Learners appreciate the utility of mathematics in real-life design and manufacturing contexts (such as fabricating tea-collection containers for Kericho farmers); demonstrate persistence when experimental results do not match theory; and show respect for peers' contributions during group construction and discussion.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | Across the first six lessons learners unpacked why the cylindrical drum holds exactly three times the conical cover ( $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$ for equal base and height), derived the frustum formula by removing a smaller cone from a larger one, and applied all three formulae to real containers. In Lesson 7 learners now become the designers: they build their own scale-model versions of the three Kericho containers, take real measurements, compute theoretical capacities, and test them experimentally — transforming abstract formulae into tangible, testable objects and bringing the driving question one step from a full answer. |
| <b>Safety Notes</b>         | Scissors and craft knives must only be used under teacher supervision and always directed away from the body. Water used for filling models must be contained in a tray or basin to prevent slipping hazards on the floor. Learners with cuts or open wounds should wear gloves or assign measuring/recording roles instead of water-handling roles. All waste cardboard offcuts must be collected and placed in a bin — do not litter the classroom.                                                                                                                                                                                                                          |

### B. LESSON OVERVIEW

Learners work in groups of four to construct three model containers from cardboard: (1) a cone, (2) a frustum of a cone, and (3) a frustum of a square pyramid. Each group uses cereal-box cardboard, a ruler, compass, protractor, scissors, and tape. After construction, they measure the relevant dimensions of each model, apply the appropriate volume formula to calculate theoretical capacity in millilitres/litres, then fill the model with water or fine dry sand measured in a graduated cylinder to obtain an experimental capacity. Groups record both values, compute the percentage discrepancy, and discuss possible sources of error. The lesson culminates in a gallery display of models and a whole-class discussion connecting measured discrepancies back to the phenomenon: just as real Kericho tea-collection containers have manufacturing tolerances, our models show that small errors in dimension produce measurable errors in capacity — reinforcing why precise measurement and correct formula choice matter.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                             | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Volumes of cones intuition</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Volume of the Cone (PLIX)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Each group receives three pre-drawn net templates (cone, frustum of cone, frustum of square pyramid) on plain paper with blank dimension labels. Before cutting anything, learners predict — individually in their exercise books — which of the three finished models will have the greatest capacity, which the least, and by roughly how much. They also estimate the dimensions they will try to achieve (e.g., 'I think our cone will be about 10 cm tall and hold about 100 mL'). Predictions are written using the sentence frame: 'I predict the [shape] will hold the most because...' | Display the three blank net templates on the board. Say: 'Before we build anything, I want every one of you to be a scientist — make a prediction you can test. Look at these three nets. Which finished container do you think will hold the most water? Which will hold the least? Write your prediction AND your reason in your exercise book right now. You have 3 minutes — go.' [Allow 3 minutes of silent individual writing.] After 3 minutes: 'Turn to your neighbour and share your prediction in 30 seconds each.' [Allow 1 minute pair-share.] Cold-call 3 learners — choose one from a high-confidence hand-raiser, one from a non-volunteer, one from a different gender. Ask each: 'What did you predict, and what is your mathematical reason?' [WAIT TIME: 12 seconds after each question before prompting.] Record the three predictions on the board in a T-chart labelled 'Our Prediction' and 'Our Reason'. Say: 'We will return to these predictions at the end of the lesson. Keep them honest — do not change them once we start building.' | Prediction Journaling with a Sentence Frame — activates prior knowledge of the three volume formulae from Lessons 3–6 and creates cognitive commitment that will be productively disrupted when experimental results arrive. Writing before sharing prevents groupthink and gives the teacher rapid formative data on which formulae learners can recall.        | Circulate during the 3-minute writing window and scan exercise books. Observable indicator of readiness: learner has written a specific numerical estimate (not just 'the cylinder holds more') and referenced at least one dimension or formula variable (e.g., 'because $V = \frac{1}{3}\pi r^2 h$ so a cone with the same base as the frustum will hold less'). If fewer than half the class can produce a formula reference, pause and do a 2-minute choral recall of the three formulae on the board before proceeding. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Volume of a cone</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Volume of</a>                                                                                                                                                                            | Groups construct their three cardboard models following the net templates. The cone net requires learners to calculate the slant length $l = \sqrt{h^2 + r^2}$ to draw the correct sector. The frustum-of-cone net requires two concentric circles (radii $R$ and $r$ for the base and top) plus the lateral curved band. The frustum-of-pyramid net requires four trapezoidal faces                                                                                                                                                                                                            | Distribute materials (cardboard sheets, rulers, compasses, scissors, tape, basin of water, 250 mL graduated cylinders). Say: 'Each group will build all three models. Before you cut, the measurer in your group must write the target dimensions on the net in pencil — agree on them as a group. For the cone, decide: what height $h$ and base radius $r$ do you                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hands-On Model Construction with Targeted Measurement Recording — learners experience the physical reality that 'h' in the formula is the perpendicular (vertical) height, not the slant height, a distinction that is easy to overlook when working only on paper. Requiring measurement of the actual built model (which may differ from the target) makes the | Use an observation checklist with three checkpoints: (1) Group has written target dimensions before cutting — yes/no; (2) Group correctly identifies perpendicular height (not slant height) when prompted — yes/no; (3) Group records both experimental trial readings and computes an average — yes/no. If a group confuses slant height with                                                                                                                                                                              |

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| <p><a href="#">Pyramids: Fluctuating Height (PLIX)</a><br/>Source: CK-12<br/><a href="#">Search ARES</a><br/>for similar readings</p>                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>plus top and bottom squares. After cutting and taping, each group measures: (a) for cone — base radius <math>r</math> and perpendicular height <math>h</math>; (b) for frustum of cone — bottom radius <math>R</math>, top radius <math>r</math>, and perpendicular height <math>h</math>; (c) for frustum of pyramid — bottom side length <math>a</math>, top side length <math>b</math>, and perpendicular height <math>h</math>. All measurements are recorded in a group results table drawn in exercise books. Groups then fill each model with water (or dry sand) using a graduated cylinder/measuring cup and record the experimental volume in mL.</p>                                                                             | <p>want? For the frustum of a cone, decide <math>R</math>, <math>r</math>, and <math>h</math>. For the pyramid frustum, decide <math>a</math>, <math>b</math>, and <math>h</math>. Write these target dimensions down before you cut anything.' [Allow 5 minutes for planning and marking dimensions.] Circulate — prompt groups who have not yet written target dimensions: 'Tell me your target <math>R</math>, <math>r</math>, and <math>h</math> for the frustum before you cut.' After models are built (approximately 25 minutes of construction): 'Now measure your ACTUAL dimensions — the model may not be exactly what you planned. Measure three times and take the average.' [WAIT TIME: allow 5 minutes for measurement.] When filling with water: 'Use the graduated cylinder. Fill slowly. Record to the nearest 5 mL. Do this twice and average the two readings.' Cold-call 2 groups mid-construction to share one challenge they have encountered: [WAIT TIME: 12 seconds] 'Group 3 — what is the hardest part of making the frustum of cone net? Why?' and 'Group 5 — how did you work out the slant height for your cone net?'</p> | <p>connection between real-world manufacturing tolerance and mathematical precision explicit and observable.</p>                                                                                                                                                                                                                                                                                                                                                                                             | <p>perpendicular height during measurement, intervene immediately: 'Show me where the perpendicular height is on this frustum. Drop an imaginary plumb-line from the centre of the top face straight down — that distance is <math>h</math>. Measure that, not the slant side.'</p>                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Volumes of cones intuition</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a><br/>for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Volume of prisms and pyramids</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a><br/>for similar</p> | <p>Using their actual measured dimensions, each group calculates theoretical volume for all three models: <math>V_{\text{cone}} = \frac{1}{3}\pi r^2 h</math>; <math>V_{\text{frustum-cone}} = \frac{1}{3}\pi h(R^2 + Rr + r^2)</math>; <math>V_{\text{frustum-pyramid}} = \frac{1}{3}h(a^2 + b^2 + ab)</math>. They convert volumes from <math>\text{cm}^3</math> to mL (<math>1 \text{ cm}^3 = 1 \text{ mL}</math>) and to litres. They then compute the percentage discrepancy: <math>\% \text{ discrepancy} =  (\text{theoretical} - \text{experimental})  / \text{theoretical} \times 100\%</math>. Each group discusses and writes at least two reasons why their experimental capacity differed from theory (e.g., gaps/overlaps in</p> | <p>Project (or write on the board) the three formulae with units clearly labelled. Say: 'Now comes the mathematics. Use your ACTUAL measured dimensions — not your targets — to calculate the theoretical volume in <math>\text{cm}^3</math>. Then convert to mL and to litres. Then compare with your experimental reading and work out the percentage discrepancy. Show every step.' [Allow 10 minutes for calculation.] Circulate and check formula substitution — specifically watch for learners squaring only one radius term in the frustum</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Discrepancy Analysis with Phenomenon Re-connection — rather than treating experimental error as a failure, learners are guided to see it as a mathematical insight: the frustum formula is sensitive to small changes in <math>R</math>, <math>r</math>, and <math>h</math> simultaneously, which explains why the Kericho metal bucket's capacity is so different from the cone despite similar outer appearance. This is the 'Explain' of the POE cycle and directly advances the driving question.</p> | <p>Collect each group's written calculation page. Observable indicators: (1) Correct formula stated with all variables labelled; (2) Units consistently in cm and <math>\text{cm}^3</math> throughout before conversion; (3) Percentage discrepancy computed correctly using the formula; (4) At least two physically plausible sources of error identified. Flag any group whose theoretical volume is more than 40% away from the experimental — this almost always indicates that slant height was used instead of perpendicular height, and</p> |

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| <p><a href="#">readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>cardboard seams, irregular base, parallax when reading the graduated cylinder, the cardboard itself displacing volume). Groups then connect this explicitly to the Kericho phenomenon: 'If a metalworker in Kericho makes a bucket that is 1 cm shorter than the design, how much capacity does the farmer lose?'</p>                                                                                                                                                                                                                                                                                                             | <p>formula. If spotted, ask: 'Read the formula aloud. How many radius terms are there? Which ones did you square?' After calculations: facilitate a 5-minute whole-class share. Cold-call 4 groups — one per model type plus one additional: 'Group 2 — what percentage discrepancy did you get for your cone, and what do you think caused it?' [WAIT TIME: 12 seconds.] Write all groups' percentage discrepancies on the board in a class results table. Ask: 'Look at the discrepancies for the frustum of cone across all groups. Are they bigger or smaller than for the cone? Why might that be?' [WAIT TIME: 15 seconds, then take 2 hands.] Finally: 'Now think about the Kericho bucket. If Achieng' the metalworker makes the bucket 1 cm shorter — h decreases by 1 cm — use your formula to estimate how many fewer millilitres the farmer can carry. Calculate it.' [Allow 3 minutes.]</p> |                                                                                                                                                                                                                                                                                                                                                                                | <p>requires immediate individual re-teaching.</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a cone</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Volume of Cones (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Each group writes one new sticky note (or index card) to add to the class Driving Question Board. The note must complete one of two sentence starters: (A) 'We can now explain... [part of the driving question]' — for something this lesson has answered; or (B) 'We still need to find out... [remaining gap]' — for something unresolved. Groups also vote on which sticky note from previous lessons (Lessons 1–6) can now be moved from the 'Questions' column to the 'Explained' column based on today's evidence. The teacher leads a 3-minute whole-class review of the DQB, noting what has been explained and what</p> | <p>Direct learners to the DQB display. Say: 'You have 4 minutes as a group to write your DQB card. Choose sentence starter A if you can now fully explain something about our Kericho farmers' containers. Choose starter B if today's lesson raised a new question you want answered.' [Allow 4 minutes.] Invite one representative from each group to place their card on the board and read it aloud in one sentence. [WAIT TIME: 10 seconds between each group.] Then say: 'Scan the old cards from Lessons 1 to 6. Raise your hand if you see a question card that today's project has fully answered — we can</p>                                                                                                                                                                                                                                                                                  | <p>Cumulative DQB Curation — making learners explicitly classify their new knowledge as 'explained' versus 'still open' builds metacognitive awareness and reinforces that the driving question is answered progressively, not all at once. Moving old question cards to the Explained column gives a visible, motivating record of intellectual progress across the unit.</p> | <p>Read all new DQB cards after the lesson. Observable indicator of understanding: the 'We can now explain' card must reference a specific formula, a specific shape (cone/frustum), or the experimental verification process — not a vague statement like 'we know more about volume.' Cards that are vague are returned to the group at the start of Lesson 8 with the prompt: 'Add one formula or measurement to make this more specific.'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | remains for Lesson 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | move it to the Explained column.' Cold-call 2 volunteers to nominate a card and explain why it can be moved. [WAIT TIME: 12 seconds per volunteer.] Summarise: 'Here is where we stand. We can now explain why the Kericho cone, bucket and drum hold different amounts — we have the formulae AND we have built and tested physical models. What remains for Lesson 8 is...' [preview the capstone lesson].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Visually Comparing the Capacity and Volume of Two Containers (Metric Units)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | Groups arrange their three finished models on their desk as a mini-gallery, with a hand-written label card for each model showing: shape name, actual measured dimensions, theoretical volume ( $\text{cm}^3$ and litres), experimental volume (mL), and percentage discrepancy. Each group also produces a one-page 'Design Sheet' in their exercise book: a labelled diagram of each model with dimensions, the formula used, the full calculation, and a one-sentence reflection on the biggest source of error. At least one learner in each group photographs the three models together alongside the label cards using a phone or tablet. The teacher collects one photograph per group (or accepts a printed image) as the documentary evidence for the project assessment. The lesson closes with learners returning to their original predictions from Phase 1 and writing a two-sentence reflection: 'My prediction was [correct/partially correct/incorrect] because...' | Say: 'You have 8 minutes to write your label cards and complete your Design Sheet. The label card must show six things — I will write them on the board.' [Write: 1. Shape name. 2. Measured dimensions. 3. Theoretical volume in $\text{cm}^3$ . 4. Theoretical capacity in litres. 5. Experimental volume in mL. 6. % discrepancy.] Circulate and prompt any group that has left a label blank: 'You have a blank space for percentage discrepancy — finish that calculation before you display.' After 8 minutes: 'Take your photograph now. Make sure all three models and all three label cards are visible in the frame.' Conduct a 3-minute gallery walk — ask learners to silently look at two other groups' models and label cards. Then say: 'Back to your seats. Open your exercise book to your prediction from the very start of today's lesson. Write two sentences: was your prediction correct, and what does the mathematics tell you now?' Cold-call 2 learners to share their reflection aloud. [WAIT TIME: 12 seconds each.] Close by saying: 'The Kericho farmers' bucket looks similar to the cone and the drum, but today you proved with your | Gallery Display with Prediction Revisit — anchoring the physical model display to the original written prediction closes the POE (Predict–Observe–Explain) cycle explicitly. The two-sentence reflection requires learners to reconcile their intuition with the mathematical evidence, consolidating the conceptual shift from 'shapes that look similar hold similar amounts' to 'volume depends precisely on the formula, and small dimensional changes produce measurable capacity differences.' | Collect the Design Sheet (one per learner, drawn in exercise book). Observable indicators: (1) All three models have a correctly labelled diagram with at least three dimensions marked; (2) The correct formula is stated and correctly applied for each shape; (3) Percentage discrepancy is calculated and a physically plausible error source is named; (4) The prediction reflection is specific and references the formula or a measured dimension. Use this as the project formative rating against the KICD rubric descriptor: 'The learner accurately determines the volume and capacity of cones, frustum of a cone, and frustum of a pyramid.' |

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|  |  | own hands and your own calculations that the mathematics explains every difference in capacity. In Lesson 8 we will use everything we have built and proved to answer our driving question fully and present our solution.' |  |  |
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did all groups successfully identify perpendicular height as distinct from slant height during measurement? If not, this misunderstanding must be explicitly corrected at the start of Lesson 8 before any capstone calculations. (2) What was the typical percentage discrepancy across the class? A class median discrepancy above 20% suggests that cardboard construction introduced too much leakage/irregularity — for a future iteration, consider sealing model bases with masking tape or using plastic cups as the frustum base to reduce water loss. (3) Were the DQB 'We can now explain' cards specific enough to demonstrate formula-level understanding? If more than a third of the class produced vague cards, build a 5-minute DQB re-visit into the opening of Lesson 8. (4) Did the prediction-revisit reflection reveal lingering misconceptions — for example, learners who still attribute the difference in capacity to the 'opening size' rather than the full three-dimensional formula? These learners need a targeted follow-up question at the start of Lesson 8: 'If I keep the same top and bottom radius but double the height of the Kericho bucket, how does the volume change? Use the formula to show me.' (5) Consider which group's model set would make the strongest anchor display for Lesson 8's capstone — select the group with the most precise construction (lowest % discrepancy) AND the group with the highest discrepancy, so learners can compare and discuss why precision matters in real container design.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| What did I observe?                                          | Learners built three cardboard model containers (a cone, a frustum of a cone, and a frustum of a square pyramid), measured their actual dimensions, and filled them with water to determine experimental capacity — observing directly that the three shapes, even when built to similar heights and base widths, hold measurably different amounts of water, mirroring the Kericho farmers' containers.                                                                                                                 |
| What did I learn?                                            | Learners applied $V = \frac{1}{3}\pi r^2 h$ for the cone, $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ for the frustum of a cone, and $V = \frac{1}{3}h(a^2 + b^2 + ab)$ for the frustum of a pyramid to compute theoretical capacity in $\text{cm}^3$ and litres; they converted between $\text{cm}^3$ and mL ( $1 \text{ cm}^3 = 1 \text{ mL}$ ); and they quantified the difference between theoretical and experimental capacity as a percentage discrepancy, identifying measurement and construction error as the cause. |
| How does this explain the phenomenon?                        | Learners can now explain that the Kericho tea farmers' cone, bucket (frustum of cone), and cylindrical drum hold such different amounts not because of how similar they look on the outside, but because their internal three-dimensional shapes obey fundamentally different volume formulae — and that even a 1 cm error in manufacturing a container's height or radius produces a calculable and significant change in the amount of tea it can carry.                                                               |



## LESSON 8 (80 minutes): Model Building Culmination, Final Explanation & Assessment: The Kericho Farmers' Containers Solved

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Purpose</b>              | To consolidate all five specific learning outcomes for Sub-Strand 2.7 by presenting project models, completing the Driving Question Board, constructing a final annotated volume-relationship diagram, and demonstrating mastery through a written class activity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Knowledge</b>            | Learners recall and integrate: (a) the formula for volume/capacity of a cone and pyramid; (b) the one-third relationship between a cone and its enclosing cylinder (and pyramid and its enclosing prism); (c) the frustum formula derived by subtraction of cones/pyramids; (d) unit conversions between $\text{cm}^3$ , litres and millilitres; (e) sources of measurement error in physical models.                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Skills</b>               | Learners are able to: (a) determine the volume and capacity of cones and pyramids; (b) compute volume and capacity of a frustum of a cone; (c) calculate the volume and capacity of a frustum of a pyramid; (d) apply volume and capacity reasoning to design a container of specified capacity; (e) communicate mathematical reasoning through annotated diagrams and written explanations.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Attitudes</b>            | Learners appreciate the utility and aesthetic value of volume and capacity mathematics in everyday Kenyan life; demonstrate intellectual honesty when comparing predicted versus calculated values and explaining sources of error; show respect for peers' project work during gallery presentations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Key Inquiry Question</b> | Why is the knowledge of volume and capacity useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose in Storyline</b> | This is the culminating lesson of the unit. Learners have, across Lessons 1–7, built understanding of cylinder, cone, pyramid and frustum volumes. In Lesson 8 they bring everything together: groups present their frustum-container project models, compare calculated and measured capacities, and explain discrepancies. The whole class then constructs a final annotated diagram that captures the complete volume-relationship story — from the Kericho farmers' cylindrical drum, conical sisal cover and metal bucket (frustum) encountered in Lesson 1, to the full algebraic derivations. The DQB columns are completed. A written class activity formally assesses all five SLOs. Learners articulate what they now know versus what they predicted in Lesson 1, closing the anchoring phenomenon loop. |
| <b>Safety Notes</b>         | When handling physical containers filled with water during project demonstrations, ensure floor surfaces are kept dry to prevent slipping. Learners should place containers on a tray or basin. No sharp metallic edges on homemade models should be left exposed — tape or file all cut edges before the presentation session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### B. LESSON OVERVIEW

Lesson 8 is the 80-minute culminating session for Sub-Strand 2.7. It opens with a brief Predict phase in which learners revisit their Lesson 1 predictions about the Kericho farmers' three containers and privately write what they now believe. Groups then present their frustum-container project models (Observe/Evidence phase) in a structured gallery walk, sharing both calculated capacities and physically measured volumes using water. In the Explain phase, the teacher facilitates whole-class construction of a final annotated diagram showing the volume relationships:  $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$  (same base and height);  $V_{\text{frustum}} = V_{\text{big\_cone}} - V_{\text{small\_cone}}$ ; and the parallel relationships for pyramids and their frustums. The Driving Question Board is completed collaboratively, with learners moving evidence cards to the 'Answered' column and articulating remaining uncertainties. The lesson closes with a 25-minute written class activity assessing all five SLOs, followed by a 5-minute teacher-led debrief connecting every answer back to the Kericho phenomenon.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                    | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Volumes of cones intuition</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a><br>for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Surface Area and Volume of Solids Practice</a><br>Source: CK-12<br> <a href="#">Search ARES</a><br>for similar readings | Each learner opens their exercise book to the prediction entry they made in Lesson 1 about the Kericho farmers' three containers. They silently re-read their original prediction, then write a 3-sentence 'revised prediction' in a different colour: (1) what they now believe about WHY the cylinder holds three times as much as the cone; (2) how much MORE the frustum bucket holds than the cone; and (3) what mathematical tool they would use to design a frustum that holds exactly 20 litres. Learners then share with their shoulder partner (Think-Pair) before two pairs are cold-called to share with the class. | Display the anchoring phenomenon photograph (or sketch on the board) of the three Kericho containers side by side — the cylindrical drum, the conical sisal cover, and the metal frustum bucket — labelled with their shared height ( $h = 30$ cm) and shared opening diameter ( $d = 28$ cm). Say verbatim: 'In Lesson 1 you made a prediction about these three containers. Before we present our projects, I want you to meet your Lesson 1 self. Open your books to that page.' Allow 3 minutes of silent individual writing. Then say: 'Share your revised prediction with your shoulder partner. You have 90 seconds.' After pair sharing, cold-call exactly 2 pairs using the class register (row 3, seat 2; row 5, seat 4). For each response, probe: 'What evidence from our lessons made you change your mind?' Apply 12-second WAIT TIME before accepting the answer. Record key phrases from responses on the board under the heading 'What we NOW believe' — this will anchor the DQB completion in Phase 4. | Predict–Revisit Strategy: By returning to their original Lesson 1 written prediction and comparing it against current understanding, learners engage in metacognitive reflection — a core sensemaking move. Seeing their own earlier misconceptions in writing makes conceptual change visible and personally meaningful, motivating the final explanation phase.                                                                                       | Observable indicator: Each learner's revised prediction explicitly names the $V = (1/3)\pi r^2 h$ relationship for the cone and references the frustum subtraction formula. Teacher circulates during the 3-minute writing window and scans for learners who still write vague answers (e.g., 'because the shapes are different') — these learners are flagged and will be specifically monitored during the written assessment in Phase 5.  |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Visually Comparing the Capacity and Volume of Three Containers (Metric Units)</a><br>Source: CK-12<br> <a href="#">Search ARES</a><br>for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Surface Area and</a>                                                                                                                                                             | Each of the 5–6 project groups displays their frustum-container model on their desk alongside: (a) a completed project data sheet showing the measured dimensions ( $R$ , $r$ , $h$ ), their calculated volume in $\text{cm}^3$ and capacity in litres, and the target capacity they were designing for; (b) a 250 ml graduated measuring cylinder or calibrated bottle used to physically measure the water volume their container actually holds. During a 12-minute structured gallery walk,                                                                                                                                 | Before the gallery walk begins, distribute the Gallery Observation Sheet and model how to calculate percentage error: write on the board '% error = $ \text{calculated} - \text{measured}  \div \text{calculated} \times 100\%$ ' and work through one fictional example: 'Group Simba calculated 18.4 L but measured 17.1 L. What is the % error? Work it out — you have 30 seconds.' Cold-call 1 learner. Confirm: ' $(18.4 - 17.1) \div 18.4 \times 100\% \approx 7.1\%$ .' During the gallery walk, circulate and pose to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Evidence-to-Explanation Gallery Walk: Learners physically interact with multiple peer models, compute percentage error, and hear groups articulate their own sources of discrepancy. This data-rich observation phase provides concrete, multi-group evidence that the frustum formula is consistently predictive, while also surfacing the real-world gap between mathematical calculation and physical measurement — a key scientific and engineering | Observable indicator: Each learner's completed Gallery Observation Sheet contains correct percentage error calculations for at least 4 groups. Groups' verbal summaries explicitly name a geometric source of error (e.g., 'the walls of our model were not perfectly straight so the top radius was slightly smaller than we measured') rather than a vague one (e.g., 'we made a mistake'). Teacher collects two randomly selected Gallery |

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| <p><a href="#">Volume of Cones Practice</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>learners rotate through every group's display in 2-minute slots, using a Gallery Observation Sheet to record: Group name   Target capacity (L)   Calculated capacity (L)   Measured capacity (L)   % error. In the final 8 minutes, each group stands by their model and delivers a 90-second verbal summary: 'Our container is a frustum of a cone / pyramid. We designed it to hold ____ litres. Our formula gave ____ litres. When we filled it with water it held ____ litres. The difference is because ____.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>individual learners: 'Point to R and r on this model. Which one is bigger and why?' and 'If this group made R and r both larger by the same amount, would the volume increase, decrease or stay the same — explain.' Apply 12-second WAIT TIME at each probe. After the group verbal summaries, ask the whole class: 'Which group had the smallest percentage error? What did they do differently in construction?' Cold-call 3 learners for responses.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>practice (Planning and Carrying Out Investigations; Analysing and Interpreting Data).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Observation Sheets from different rows to check calculation accuracy.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Comparing the Capacity and Volume of Three Containers (Metric Units)</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Surface Area and Volume of Pyramids Practice</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p> | <p>Learners work in their project groups to construct a Final Annotated Diagram in their exercise books. The diagram must show all four solids studied — cylinder, cone, square pyramid, frustum of cone — drawn to roughly the same height and base, each labelled with its volume formula. Arrows between shapes must be annotated with the mathematical relationship: a double-headed arrow between cylinder and cone labelled '<math>V_{\text{cylinder}} = 3 \times V_{\text{cone}}</math> (same base, same height)'; an arrow from frustum to big cone and small cone labelled '<math>V_{\text{frustum}} = V_{\text{big}} - V_{\text{small}} = (1/3)\pi h(R^2 + Rr + r^2)</math>'. At the bottom of the diagram, learners write one sentence connecting each shape back to the Kericho phenomenon: e.g., 'The cylindrical drum holds <math>3 \times</math> the conical sisal cover because they share the same base diameter and height.' Groups then share their diagram with an adjacent group and give one piece of written feedback on a sticky note.</p> | <p>Begin by drawing the outline of the four shapes on the board — large enough for learners to see the proportional size differences. Say verbatim: 'I am going to draw the skeleton. Your job is to fill in every formula, every relationship arrow, and every connection to the Kericho containers. You have 10 minutes in your groups.' Set a visible timer. After 10 minutes, cold-call one group to come to the board and annotate the class diagram, narrating each step aloud. After they annotate the cone–cylinder relationship, ask: 'Can someone from a different group give me the frustum formula and explain where it comes from — in one sentence?' Apply 12-second WAIT TIME. Cold-call 2 learners. Then ask: 'What is the capacity in litres of a cone with the same dimensions as the Kericho cylindrical drum — radius 14 cm, height 30 cm?' Allow 90 seconds of individual working. Cold-call 1 learner: expected answer <math>V_{\text{cone}} = (1/3) \times \pi \times 14^2 \times 30 \approx 6,158 \text{ cm}^3 \approx 6.16 \text{ L}</math>. Confirm and say: 'Now you understand why the conical sisal cover holds only about 6 litres</p> | <p>Annotated Diagram Synthesis (Science and Engineering Practice: Developing and Using Models): Constructing a diagram that simultaneously displays all four formulae and their inter-relationships forces learners to organise knowledge into a coherent relational structure — moving from isolated procedural knowledge (calculating one volume at a time) to integrated conceptual understanding (how all the shapes relate through the one-third ratio and the subtraction principle). Connecting each arrow back to the Kericho phenomenon ensures the model is anchored in evidence rather than being purely abstract.</p> | <p>Observable indicator: Learner diagrams contain (a) all four correct volume formulae with no algebraic errors; (b) the annotated <math>V_{\text{cylinder}} = 3 \times V_{\text{cone}}</math> arrow; (c) the frustum = big cone – small cone derivation arrow; (d) at least one Kericho-phenomenon sentence per shape. Teacher checks 4–5 books during group work. Groups that omit the pyramid frustum formula are prompted: 'How is the frustum of a pyramid different from the frustum of a cone? Write the formula and add the arrow.'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | while the cylindrical drum holds about 18.5 litres — and the frustum bucket sits between them depending on its r.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a cone</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Surface Area and Volume of Pyramids (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>The class DQB (displayed on the wall or front board since Lesson 1) has three columns: 'What we wonder', 'Evidence we have gathered', 'What we now know/can explain'. Learners spend 3 minutes in pairs reviewing all cards in the 'What we wonder' column and deciding which questions have now been fully answered. A class representative physically moves answered cards to the 'What we now know' column and writes a one-line answer on the card. Any question that remains unanswered or only partially answered is moved to a new column: 'Questions for further study'. The final 2 minutes involve each learner writing in their book: 'In Lesson 1 I predicted _____. I now know _____ because _____.'</p> | <p>Stand at the DQB and read aloud the original Driving Question: 'Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?' Then say verbatim: 'This was our question for eight lessons. Look at this board. In pairs, decide — have we answered it? Give me a thumbs up when your pair agrees.' Wait until at least 80% show thumbs up (approximately 10 seconds). Ask one pair: 'State the complete answer in two sentences.' After the class representative moves cards, teacher reads aloud the final answer card for the main driving question. Ask: 'Is there anything on this board we could NOT fully answer? What further mathematics would we need?' Accept 2 responses. This surfaces intellectual humility and models scientific practice of identifying the boundaries of current knowledge.</p> | <p>Driving Question Board Completion (Science and Engineering Practice: Constructing Explanations): The DQB is a public, cumulative knowledge-tracking tool. Physically moving cards from 'wonder' to 'know' makes the arc of learning tangible and celebratory, while identifying residual questions models authentic scientific inquiry — real understanding always generates new questions. Comparing Lesson 1 predictions with final answers makes conceptual growth explicit.</p> | <p>Observable indicator: Each learner's written 'In Lesson 1 I predicted ____ / I now know ____' entry explicitly references at least one formula or mathematical relationship (not just a general statement). Teacher scans 6–8 books during the 2-minute writing window. Entries that remain at the level of 'I now know the shapes are different' are flagged as needing support during the written assessment.</p>                                                                                                                                               |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a Pyramid - Overview</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b></p>                                                                                                                                                                                                                                                          | <p>Learners individually complete a structured written class activity (20 minutes) comprising five questions, one per SLO, all set in the Kericho tea-farming context. Q1 (SLO a — cone/pyramid): A conical sisal cover used by a Kericho farmer has a base diameter of 56 cm and a vertical height of 30 cm. Calculate its volume in cm<sup>3</sup> and capacity in litres. Q2 (SLO b — frustum of</p>                                                                                                                                                                                                                                                                                                                  | <p>Distribute the written activity sheet face-down. Say verbatim: 'This activity covers everything we have learned in Sub-Strand 2.7. Every question is set in Kericho — these are real containers used by real farmers. Show all working. You have 20 minutes starting now.' Set timer visibly. Circulate quietly for the first 5 minutes — do not answer questions; instead, if a learner is stuck, point silently to</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Authentic Assessment as Final Sensemaking (Science and Engineering Practice: Using Mathematics and Computational Thinking): The five-question written activity is not merely summative — it is the final sensemaking act of the unit. By requiring learners to apply every SLO within the single coherent context of the Kericho phenomenon, the assessment</p>                                                                                                                     | <p>Observable indicators during debrief: (a) Q1 — learner correctly halves the diameter to get <math>r = 28</math> cm before substituting; (b) Q2 — learner correctly expands <math>R^2 + Rr + r^2 = 196 + 126 + 81 = 403</math>; (c) Q3 — learner uses the pyramid frustum formula with <math>A_1 = 400</math> cm<sup>2</sup>, <math>A_2 = 144</math> cm<sup>2</sup>, <math>\sqrt{(A_1 A_2)} = 240</math> cm<sup>2</sup>, giving <math>V = (15/3)(400 + 144 + 240) = 5 \times 784 = 3,920</math> cm<sup>3</sup>; (d) Q4 — learner sets up <math>10,000 =</math></p> |

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| <p><a href="#">Volume of the Cone (PLIX)</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p> | <p>cone): The metal bucket (frustum of cone) has a large base radius <math>R = 14</math> cm, small base radius <math>r = 9</math> cm and height <math>h = 30</math> cm. Use the formula <math>V = (1/3)\pi h(R^2 + Rr + r^2)</math> to calculate its volume in <math>\text{cm}^3</math> and capacity in litres. Q3 (SLO c — frustum of pyramid): A square frustum used as a tea-drying tray has a large base of side 20 cm, a small base of side 12 cm and height 15 cm. Calculate its volume. Q4 (SLO d — application/design): A farmer in Kericho needs a frustum-of-cone container that holds exactly 10 litres (10,000 <math>\text{cm}^3</math>). She fixes <math>h = 25</math> cm and <math>r = 6</math> cm. Use the frustum formula to find <math>R</math>, correct to 1 decimal place. Q5 (SLO e — appreciation/explanation): In 3–4 sentences, explain why the cylindrical drum, conical cover and frustum bucket all have the same opening diameter and height yet hold different amounts. Use the formulae to support your answer. After 20 minutes, teacher leads a 5-minute oral debrief: cold-calls learners for each answer and connects every answer back to the Kericho phenomenon.</p> | <p>the formula sheet displayed on the board (all four formulae written: <math>V_{\text{cone}} = (1/3)\pi r^2 h</math>; <math>V_{\text{pyramid}} = (1/3)Ah</math>; <math>V_{\text{frustum\_cone}} = (1/3)\pi h(R^2 + Rr + r^2)</math>; <math>V_{\text{frustum\_pyramid}} = (h/3)(A_1 + A_2 + \sqrt{A_1 A_2})</math>). At the 15-minute mark, give a 5-minute warning. After collection, debrief Q1 by cold-calling 1 learner (expected: <math>V = (1/3) \times \pi \times 28^2 \times 30 \approx 24,630 \text{ cm}^3 \approx 24.63 \text{ L}</math> — note: diameter 56 cm means <math>r = 28</math> cm, not 14 cm, so cone is much larger; confirm with class). Debrief Q2: cold-call 1 learner (expected: <math>V = (1/3) \times \pi \times 30 \times (196 + 126 + 81) = (1/3) \times \pi \times 30 \times 403 \approx 12,662 \text{ cm}^3 \approx 12.66 \text{ L}</math>). For Q5, cold-call 1 learner and require them to read their answer aloud; prompt: 'Which word in your answer shows you understand the ONE-THIRD relationship?' Close by saying: 'These are the containers of the Kericho tea farmers. You can now calculate exactly how much water — or tea — any of them holds. That is why the knowledge of volume and capacity is useful.'</p> | <p>reinforces that the formulae are tools for explaining a real situation, not isolated procedures. The oral debrief after the written activity closes the explanation loop publicly, making each answer a shared class conclusion rather than a private grade.</p> | <p><math>(1/3)\pi \times 25 \times (R^2 + 6R + 36)</math> and solves to get <math>R \approx 11.4</math> cm; (e) Q5 — answer explicitly states <math>V_{\text{cylinder}} = \pi r^2 h</math> and <math>V_{\text{cone}} = (1/3)\pi r^2 h</math> so <math>V_{\text{cylinder}} = 3 \times V_{\text{cone}}</math>, and places frustum between them because its formula accumulates area between <math>R</math> and <math>r</math>. Teacher marks written activities before the next lesson and returns with written feedback aligned to the KICD four-level rubric (Below Expectations / Approaches Expectations / Meets Expectations / Exceeds Expectations) for SLO indicators in Sub-Strand 2.7.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal:

1. PHENOMENON CLOSURE: Did learners articulate a complete, formula-supported explanation of why the three Kericho containers hold different volumes, or did some still rely on shape descriptions alone? What follow-up is needed?
2. PROJECT QUALITY: What was the average percentage error between calculated and measured capacities across the project groups? Groups with errors above 15% — what construction or measurement issues caused this? How would you adjust the project brief for the next cohort?
3. DQB COMPLETION: Were there any 'What we wonder' cards that could NOT be moved to 'What we now know'? If so, which sub-strand or future grade topic



addresses them? Note these for the next unit introduction.

4. WRITTEN ASSESSMENT: After marking Q4 (the design/reverse-engineering question), what proportion of learners successfully rearranged the frustum formula to solve for R? If fewer than 60% succeeded, this algebraic manipulation skill requires reinforcement in strand 1.3 (Quadratic Equations) — note the cross-strand connection for the mathematics department meeting.
5. DIFFERENTIATION: Which learners consistently produced work at the 'Exceeds Expectations' level across phases 2–5? How will you extend them (e.g., challenge: derive the frustum formula from first principles by subtracting a small cone from a large cone algebraically)? Which learners were flagged during Phase 1 as still holding misconceptions — what targeted support will you provide?
6. KENYA CONTEXT: Did the Kericho tea-farming context feel authentic and engaging to learners, or did any group suggest a more locally relevant context for their school's county? Record suggestions for adaptation in future lessons.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe?                                          | We observed five project groups present frustum-shaped container models — some shaped as frustums of cones, others as frustums of square pyramids — each filled with water to physically measure capacity. We recorded their target capacity, calculated capacity (from the frustum formulae) and measured capacity (from water displacement), and computed the percentage error for each group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| What did I learn?                                            | We learned that: (1) a cone and a cylinder with the same base radius and height always satisfy $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$ , which is why the Kericho cylindrical drum holds exactly three times the conical sisal cover; (2) the frustum bucket's volume — $V = (1/3)\pi h(R^2 + Rr + r^2)$ — falls between those of the cone and the cylinder because it retains a wide base while tapering, accumulating more volume than the cone but less than the full cylinder; (3) the frustum formula is derived by subtracting a smaller cone from a larger cone, and the same logic applies to pyramid frustums; (4) we can reverse-engineer the formula to find an unknown dimension (such as R) when a target capacity is specified.                                                                                                                                                                                |
| How does this explain the phenomenon?                        | We can now fully explain the anchoring phenomenon: the cylindrical drum, the conical sisal cover and the metal frustum bucket look the same size because they share the same height and opening diameter, but their internal geometries — how cross-sectional area accumulates from base to top — differ fundamentally. The cylinder maintains a constant circular cross-section all the way up, so it accumulates the maximum possible volume. The cone tapers linearly to a point, cutting its volume to exactly one-third. The frustum keeps a wide bottom (radius R) and narrows only to a smaller circle (radius r, not zero), so it captures more volume than the cone but less than the cylinder. A Kericho farmer who needs a container of exactly 10 litres can use $V = (1/3)\pi h(R^2 + Rr + r^2)$ , fix h and r to match available sheet-metal sizes, and solve for R — turning mathematics into a practical design tool. |



| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |